BME302 Unit IV – Impact of Jets & Hydraulic Turbines

Lecture Notes · Fluid Mechanics and Fluid Machines

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How to use these notes. Every topic, worked example and question carries one tag. CORE must-know syllabus material at AKTU exam level: master these first. HOTS higher-order thinking: analyse, compare, justify or design (Bloom K4 to K6). ADVANCED beyond the exam level for most students: shaded orange, optional reading.

Unit overview

Unit IV covers how a water jet transfers momentum to a surface, and how that idea scales up into impulse and reaction turbines. It maps to **CO4: Mathematically analyze the flow in water pumps and turbines (K3)** and is planned for 8 lectures. Notation and standard results follow R.K. Bansal and Som, Biswas & Chakraborty.

Lecture	Topic	Key outcome
L1	Impact of jet on fixed flat and curved plates	Force from momentum equation
L2	Jet on moving plates, series of vanes, unsymmetrical curved vane	Work, efficiency, vane angles
L3	Classification of turbines, heads, efficiencies, Euler equation	General work equation
L4	Pelton wheel: construction, velocity triangles, work and efficiency	Maximum efficiency condition
L5	Pelton design relations, governing, numericals	Size a Pelton wheel
L6	Francis turbine: construction, draft tube, velocity triangles	Inlet/outlet vane angles
L7	Kaplan turbine: construction, velocity triangles, numericals	Axial-flow analysis
L8	Specific speed, unit quantities, selection of turbines	Choose turbine for a site

Common notation: V = absolute velocity of water, u = blade (peripheral) velocity, V_r = relative velocity, V_w = whirl (tangential) component, V_f = flow component, α = guide blade angle, θ = vane angle at inlet, ϕ = vane angle at outlet, H = net head, Q = discharge, N =

speed in rpm. Suffix 1 = inlet, 2 = outlet; η_h, η_m, η_o = hydraulic, mechanical and overall efficiency.

0. Before you start: basic ideas you need CORE

Turbine problems use four small ideas again and again: velocity triangles, power and units, momentum, and head. Read this section once before Section 1.

> **IN SIMPLE WORDS**
> A turbine is a machine that takes speed (and pressure) from water and turns it into shaft work. To follow it you only need to draw a triangle of velocities, convert units correctly, use “force equals mass per second times change of speed”, and know how a head turns into a jet speed. Each idea below comes with one tiny worked example.

0.1 Velocity triangles in simple words CORE

> **IN SIMPLE WORDS**
> Water moves, and the blade moves too. The blade “sees” the water at a different speed from the ground. Think of walking on a moving train: your speed over the ground is your walking speed plus the train speed. A velocity triangle simply draws this addition as three arrows: the water speed V over the ground, the blade speed u over the ground, and the water speed V_r as seen from the blade.

Velocities are vectors, so they add by the triangle rule. Water relative to ground = blade relative to ground + water relative to blade:

$$\vec{V} = \vec{u} + \vec{V}_r \quad \Rightarrow \quad \vec{V}_r = \vec{V} - \vec{u}$$

Symbol	Meaning	Measured	Typical value
V	Absolute velocity of water (relative to the ground)	m/s	20–80 m/s (jet)
u	Blade (peripheral) velocity	m/s	3–40 m/s
V_r	Relative velocity (water as seen from the blade)	m/s	10–40 m/s
V_w	Whirl component: part of V along the blade motion	m/s	0–50 m/s
V_f	Flow component: part of V at right angles to the blade motion	m/s	3–15 m/s
α	Angle between V and the direction of u	degrees	set by the guide vanes
θ, ϕ	Angles between V_r and the direction of u at inlet and outlet	degrees	set by the vane shape

How to draw a velocity triangle:

1. Draw $\vec{V}$ from a point, at angle α to the direction of blade motion.
2. Resolve it: $V_w = V\cos\alpha$ along the motion, $V_f = V\sin\alpha$ across it.
3. Draw $\vec{u}$ along the motion. The arrow from the tip of $\vec{u}$ to the tip of $\vec{V}$ is $\vec{V}_r$.
4. Its components are $(V_w - u)$ along the motion and V_f across it, so $\tan\theta = V_f/(V_w - u)$.

Example 0.1 CORE Water at $V = 10\ \text{m/s}$ makes 60° with the blade motion. The blade moves at $u = 3\ \text{m/s}$. Find V_r and its angle.

1. $V_w = 10\cos 60^\circ = 5\ \text{m/s}$; $V_f = 10\sin 60^\circ = 8.66\ \text{m/s}$
2. $V_r = \sqrt{(5-3)^2 + 8.66^2} = 8.89\ \text{m/s}$
3. $\tan\theta = \dfrac{8.66}{5-3} = 4.33$, so $\theta = 77.0^\circ$

0.2 Power, work, efficiency and units CORE

> **IN SIMPLE WORDS**
>
> Power is how fast work is done, in joules per second (watts). A force F pushing something that moves at speed u gives power $F \times u$. A shaft with torque T turning at ω gives power $T \times \omega$. Efficiency says how much of the power put in comes out as useful power. It is always below 100% in a real machine. Most mistakes in turbine numericals are unit mistakes, so convert first.

$$P = Fu = T\omega, \qquad \text{W.P.} = \rho g Q H, \qquad \eta = \frac{\text{power out}}{\text{power in}}$$

To convert	Rule	Example
mm to m	divide by 1000	$75\ \text{mm} = 0.075\ \text{m}$
L/s to m^3/s	divide by 1000	$700\ \text{L/s} = 0.7\ \text{m}^3/\text{s}$
rpm to rad/s	$\omega = 2\pi N/60$	$300\ \text{rpm} = 31.42\ \text{rad/s}$
rpm to blade speed	$u = \omega D/2 = \pi D N/60$	$D = 2\ \text{m}$, $N = 300$: $u = 31.42\ \text{m/s}$
W to kW	divide by 1000	$5 \times 10^6\ \text{W} = 5000\ \text{kW}$
metric hp to kW	multiply by 0.7355	$100\ \text{hp} = 73.55\ \text{kW}$

Example 0.2 CORE Water flows at $0.5\ \text{m}^3/\text{s}$ under a head of $40\ \text{m}$. Find the water power, and the shaft power if the turbine is 85% efficient.

1. $\text{W.P.} = \dfrac{\rho g Q H}{1000} = \dfrac{1000 \times 9.81 \times 0.5 \times 40}{1000} = 196.2\ \text{kW}$
2. $\text{S.P.} = 0.85 \times 196.2 = 166.8\ \text{kW}$

0.3 Impulse–momentum: force from a jet CORE

> **IN SIMPLE WORDS**
> Moving water carries momentum (mass times velocity). To change its speed or direction, something must push on it. By Newton's third law, the water pushes back on that something with an equal and opposite force. Fire hoses kick back for exactly this reason. The force equals the mass of water arriving each second times the change in its velocity.

$$F = \dot{m}\ (\ V_{\text{in}} - V_{\text{out}}\ ), \qquad \dot{m} = \rho a V$$

Here $\dot{m}$ is the mass flow rate in kg/s, a is the jet area in m^2 and V_{in}, V_{out} are the velocity components along the direction of the force. A force is the push on the water; the push of the water on the surface is equal and opposite. Sign rule: take the jet direction as positive.

Example 0.3 CORE A $25\ \text{mm}$ jet at $10\ \text{m/s}$ hits a wall and loses all its velocity in the jet direction. Find the force on the wall.

1. $a = \dfrac{\pi}{4}\ (\ 0.025\ )^{\ 2} = 4.909 \times 10^{-4}\ \text{m}^2$; $\dot{m} = 1000 \times 4.909 \times 10^{-4} \times 10 = 4.909\ \text{kg/s}$
2. $F = \dot{m}\ (\ V_{\text{in}} - V_{\text{out}}\ ) = 4.909 \times\ (\ 10 - 0\ ) = 49.1\ \text{N}$

0.4 Head, discharge and jet velocity CORE

> **IN SIMPLE WORDS**
> A height of water stores energy, as a ball held up high does. Let the water fall through a head H and it reaches speed $\sqrt{2gH}$. A nozzle turns the head into a fast jet. Real nozzles lose a little energy, so the actual jet speed is slightly smaller: this is the factor C_v (0.97–0.99). Discharge is jet area times jet speed.

$$V = C_v\sqrt{2gH}, \qquad Q = a\,V, \qquad H = \frac{p}{\rho g}$$

Example 0.4 CORE A nozzle works under a net head of $45\ \text{m}$ with $C_v = 0.98$. The jet diameter is $50\ \text{mm}$. Find the jet velocity and discharge.

1. $V = 0.98\sqrt{2 \times 9.81 \times 45} \;=\; 29.12\ \text{m/s}$
2. $Q = \dfrac{\pi}{4}\,(\,0.05\,)^{\,2} \times 29.12 =\; 0.0572\ \text{m}^3/\text{s}$

Check your understanding.

1. CORE A blade moves at $5\ \text{m/s}$ along the same line as a $12\ \text{m/s}$ jet. What is V_r? *($V_r = 12 - 5 = 7\ \text{m/s}$.)*
2. CORE Convert $600\ \text{rpm}$ to rad/s. *($\omega = 2\pi \times 600/60 = 62.83\ \text{rad/s}$.)*
3. CORE A jet carries $20\ \text{kg/s}$ at $15\ \text{m/s}$ and is brought to rest. What is the force? *($F = 20 \times 15 = 300\ \text{N}$.)*
4. HOTS With $C_v = 1$, what jet speed does a head of $20\ \text{m}$ give? *($V = \sqrt{2 \times 9.81 \times 20} \;= 19.8\ \text{m/s}$.)*

1. Hydrodynamic thrust of a jet (L1–L2) CORE

The force a jet exerts on a surface equals the rate of change of momentum of the water in that direction (impulse-momentum principle). For a jet of area a and velocity V, the mass striking per second is $\rho a V$.

$$F_x = \dot{m}\;(\,V_{1x} - V_{2x}\,) \;= \rho a V\,(\,V_{1x} - V_{2x}\,)$$

> **IN SIMPLE WORDS**
> A water jet is moving mass. When a plate stops it or turns it, the plate changes its momentum, so the plate must push on the water. The water pushes back on the plate with the same force. Think of holding your hand in the stream from a garden hose: you feel the push. Here $\rho a V$ is the kilograms of water arriving each second, and $(\,V_{1x} - V_{2x}\,)$ is how much speed each kilogram loses in the force direction.

1.1 Jet on fixed plates CORE

> **IN SIMPLE WORDS**
> A fixed plate does not move, so all the jet momentum in the force direction is lost or reversed. A flat plate only stops the jet. A curved plate turns it back, so it changes the momentum by much more. A fully reversed jet gives double the push of a stopped jet. That is why turbine buckets are cup-shaped.

Fixed flat plate held normal to the jet — water leaves parallel to the plate, so final velocity in the jet direction is zero.

$$F_x = \rho a V^2$$

Fixed flat plate inclined at θ to the jet — force normal to the plate, and its components along and across the jet:

$$F_n = \rho a V^2 \sin\theta, \quad F_x = \rho a V^2 \sin^2\theta, \quad F_y = \rho a V^2 \sin\theta\cos\theta$$

Derivation from scratch: force of a jet on a fixed flat plate (normal and inclined)

Aim. Prove $F_x = \rho a V^2$ for a normal plate and $F_n = \rho a V^2 \sin\theta$ for a plate inclined at θ to the jet.

Assumptions.

1. The jet is steady, with uniform velocity V and cross-sectional area a.
2. The plate is fixed and smooth, so it exerts no force along its own surface.
3. The jet is at atmospheric pressure everywhere, so pressure forces on the water cancel.
4. Gravity and the weight of the water are neglected.

Step 1. Mass per second. In one second a column of water of length V and area a reaches the plate. Its volume is aV, so

$$\dot{m} = \rho a V$$

Step 2. Normal plate. Take the x-axis along the jet. Each kilogram arrives with velocity V along x. It leaves sliding along the plate, at 90° to x, so its final velocity along x is zero.

$$V_{1x} = V, \qquad V_{2x} = 0$$

Step 3. Apply the impulse-momentum equation. The force of the plate on the water is the rate of change of momentum of the water:

$$F_{\text{plate on water}} = \dot{m}\ (\ V_{2x} - V_{1x}\ ) = \rho a V\ (\ 0 - V\ ) = -\rho a V^2$$

The minus sign says the plate pushes the water backwards. By Newton's third law the water pushes the plate forwards with the same magnitude:

$$F_x = \rho a V^2$$

Step 4. Inclined plate: split the velocity. Let the plate make angle θ with the jet. Resolve V into a component along the plate, $V\cos\theta$, and a component normal to the plate, $V\sin\theta$.

Step 5. Along the plate. The plate is smooth, so it cannot push the water along its surface. The momentum along the plate is therefore unchanged and gives no force.

Step 6. Normal to the plate. The normal velocity $V\sin\theta$ of the water falls to zero. The mass per second is still $\rho a V$, so

$$F_n = \rho a V \times\ (\ V\sin\theta - 0\ ) = \rho a V^2 \sin\theta$$

Step 7. Components. The normal to the plate makes angle $(\ 90^\circ - \theta\ )$ with the jet direction. Resolve F_n:

$$F_x = F_n \cos\ (\ 90^\circ - \theta\ ) = F_n \sin\theta = \rho a V^2 \sin^2\theta$$

$$F_y = F_n \sin\ (\ 90^\circ - \theta\ )\ = F_n \cos\theta = \rho a V^2 \sin\theta\cos\theta$$

Result.

$$F_x = \rho a V^2\ \ (\ \text{normal plate}\ )$$

$$F_n = \rho a V^2 \sin\theta, \quad F_x = \rho a V^2 \sin^2\theta, \quad F_y = \rho a V^2 \sin\theta\cos\theta$$

What it tells us. The force grows with the square of the velocity: doubling V gives four times the force. Putting $\theta = 90^\circ$ in the inclined result gives back the normal plate. Tilting the plate always reduces the force.

Fixed curved plate, jet striking at the centre (symmetrical) — the jet leaves at angle θ to the x-axis on both tips.

$$F_x = \rho a V^2\ (\ 1 + \ \cos\theta\ )$$

For a semicircular plate, $\theta = 0$, so $F = 2\rho a V^2$ — double the flat-plate value. This is why turbine buckets are curved.

Fixed unsymmetrical curved plate, jet striking tangentially at one tip (inlet tip angle θ, outlet tip angle ϕ):

$$F_x = \rho a V^2\ (\ \cos\theta + \ \cos\phi\ )\ , \qquad F_y = \rho a V^2\ (\ \sin\theta - \ \sin\phi\ )$$

Derivation from scratch: force of a jet on a fixed curved plate (symmetrical and unsymmetrical)

Aim. Prove $F_x = \rho a V^2\ (\ 1 + \ \cos\theta\ )$ for the symmetrical plate and $F_x = \rho a V^2\ (\ \cos\theta + \ \cos\phi\ )\ ,\ F_y = \rho a V^2\ (\ \sin\theta - \ \sin\phi\ )$ for the unsymmetrical plate.

Assumptions.

1. Steady jet of velocity V and area a at atmospheric pressure.
2. The plate is fixed and smooth, and its height change is negligible.
3. With no friction and the same pressure (atmospheric) at inlet and outlet, Bernoulli's equation shows that the water leaves with the same speed V.

Step 1. Symmetrical plate: split the jet. The jet hits the centre and divides into two equal halves. Each half has mass flow $\rho a V/2$ and leaves at angle θ to the x-axis, moving backwards.

Step 2. Velocity components of one half. The jet comes in along $+x$ and each half goes out backwards, so

$$V_{1x} = V, \qquad V_{2x} = \ -V\cos\theta$$

Step 3. Force along x. Add the two halves. Each half pushes the plate with force $\frac{\rho a V}{2}\ (\ V_{1x} - V_{2x}\ )$:

$$F_x = 2 \times \frac{\rho a V}{2}\,(V + V\cos\theta) = \rho a V^2\ (\ 1 + \ \cos\theta\ )$$

Step 4. Force across the jet. The two halves leave up and down with equal and opposite y-components, so $F_y = 0$.

Step 5. Semicircular plate. For $\theta = 0,\ \cos\theta = 1$, so $F_x = 2\rho a V^2$. This is twice the normal flat-plate force because the velocity changes from $+V$ to $-V$, not from $+V$ to 0.

Step 6. Unsymmetrical plate. Now the whole jet (mass flow $\rho a V$) enters at one tip. The inlet velocity makes angle θ with the x-axis and the outlet velocity makes angle ϕ, pointing backwards. Measure the y-components in the same sense for inlet and outlet:

$$V_{1x} = V\cos\theta, \quad V_{1y} = V\sin\theta, \qquad V_{2x} = -V\cos\phi, \quad V_{2y} = V\sin\phi$$

Step 7. Forces. Use $F = \dot{m}\,(V_1 - V_2)$ in each direction:

$$F_x = \rho a V(V\cos\theta + V\cos\phi) = \rho a V^2\,(\cos\theta + \cos\phi)$$

$$F_y = \rho a V(V\sin\theta - V\sin\phi) = \rho a V^2\,(\sin\theta - \sin\phi)$$

Result.

$$F_x = \rho a V^2\,(1 + \cos\theta) \quad (\text{symmetrical})$$

$$F_x = \rho a V^2\,(\cos\theta + \cos\phi), \quad F_y = \rho a V^2\,(\sin\theta - \sin\phi) \quad (\text{unsymmetrical})$$

What it tells us. The more the plate turns the jet back, the larger the force, up to double the flat-plate value. If $\theta = \phi$ in the unsymmetrical plate, F_y vanishes, as expected for a symmetrical shape.

1.2 Jet on moving plates CORE

IN SIMPLE WORDS

If the plate runs away from the jet, the jet hits it with less speed, so the push is smaller. A single plate runs away for ever, so only part of the water reaches it. A wheel with many plates always has one plate ready to catch the jet, so all the water is used. That is the idea behind the Pelton wheel. For such a wheel, the plate speed that gives the best use of the jet energy is half the jet speed.

When the plate moves at u in the jet direction, the jet strikes with relative velocity $(V - u)$.

Case	Force in jet direction	Work done per second	Efficiency
Single flat plate, moving	$\rho a\,(V-u)^2$	$\rho a\,(V-u)^2 u$	—
Single symmetrical curved plate, moving	$\rho a\,(V-u)^2\,(1+\cos\theta)$	$F \times u$	—
Series of flat plates (wheel)	$\rho a V(V-u)$	$\rho a V(V-u)\,u$	$2u(V-u)/V^2$, max 50% at $u = V/2$
Series of semicircular vanes	$2\rho a V(V-u)$	$2\rho a V(V-u)\,u$	$4u(V-u)/V^2$, max 100% at $u = V/2$

For a single plate only $\rho a\,(\,V-u\,)$ kg/s strikes it, because the plate keeps running away from the jet. For a series of plates on a wheel, the whole jet ρaV is used, which is the basis of the Pelton wheel.

Derivation from scratch: jet on a moving flat plate and on a series of flat plates (maximum efficiency 50%)

Aim. Prove $F=\rho a\,(\,V-u\,)^{\,2}$ for a single moving plate, and $F=\rho aV(\,V-u\,)$, $\eta=2u\,(\,V-u\,)/V^2$ with maximum 50% at $u=V/2$ for a series of plates.

Assumptions.

1. Steady jet of velocity V and area a; the plate moves at constant speed $u<V$ along the jet.
2. The plate is flat, smooth and normal to the jet.
3. For the wheel, plates follow one another so closely that no water escapes between them.

Step 1. Single plate: mass reaching the plate. In time t the jet front travels Vt but the plate moves ut in the same direction. So the length of water that actually reaches the plate in time t is $(\,V-u\,)\,t$. Per second the length caught is $(\,V-u\,)$, and the volume caught is $a\,(\,V-u\,)$:

$$\dot m=\rho a\,(\,V-u\,)$$

Step 2. Single plate: relative motion. Sit on the moving plate. The jet then arrives at speed $(\,V-u\,)$ and is stopped by the plate (it leaves sliding along the plate, so its relative velocity along the jet is zero).

Step 3. Single plate: force. Apply impulse-momentum with relative velocities:

$$F=\dot m\,[\,(\,V-u\,)\,-0]=\rho a\,(\,V-u\,)\,(\,V-u\,)\,=\rho a\,(\,V-u\,)^{\,2}$$

Step 4. Single plate: work per second. The point of application moves u metres each second:

$$\text{Work/s}=F\times u=\rho a\,(\,V-u\,)^{\,2}u$$

Step 5. Series of plates: mass used. Now the whole jet is used, so

$$\dot m=\rho aV$$

Step 6. Series of plates: velocity change. The jet arrives at V along x. After impact the water slides along the plate, so along x it moves with the plate at speed u. The change in velocity along x is $(\,V-u\,)$:

$$F=\rho aV(\,V-u\,),\qquad \text{Work/s}=Fu=\rho aV(\,V-u\,)\,u$$

Step 7. Efficiency. The kinetic energy of the jet supplied per second is $\tfrac12\dot mV^2=\tfrac12\rho aV^3$. Then

$$\eta=\frac{\rho aV(\,V-u\,)\,u}{\tfrac12\rho aV^3}=\frac{2u\,(\,V-u\,)}{V^2}$$

Step 8. Maximum efficiency: first derivative. Write $\eta = \frac{2}{V^2}\left(Vu - u^2 \right)$ and differentiate with respect to u with V constant:

$$\frac{d\eta}{du} = \frac{2}{V^2}\left(V - 2u \right) = 0 \quad \Rightarrow \quad u = \frac{V}{2}$$

Step 9. Second derivative check.

$$\frac{d^2\eta}{du^2} = -\frac{4}{V^2} < 0$$

so $u = V/2$ gives a maximum.

Step 10. Maximum value.

$$\eta_{\max} = \frac{2\left(V/2 \right)\left(V - V/2 \right)}{V^2} = \frac{2 \times \frac{V}{2} \times \frac{V}{2}}{V^2} = \frac{1}{2}$$

Result.

$$F_{\text{single}} = \rho a \left(V - u \right)^2$$

$$F_{\text{series}} = \rho a V \left(V - u \right)$$

$$\eta = \frac{2u\left(V - u \right)}{V^2}$$

$$\eta_{\max} = 50\% \text{ at } u = \frac{V}{2}$$

What it tells us. The force falls to zero as $u \to V$, and the work falls to zero as $u \to 0$, so the best power lies in between, at half the jet speed. Even then the water leaves with half of the jet energy still in it, so a flat plate wastes 50%.

Derivation from scratch: jet on moving curved vanes and on a series of semicircular vanes (maximum efficiency 100%)

Aim. Prove $F = \rho a \left(V - u \right)^2 \left(1 + \cos\theta \right)$ for a single symmetrical curved plate, and $F = 2\rho a V \left(V - u \right)$, $\eta = 4u\left(V - u \right)/V^2$, $\eta_{\max} = 100\%$ at $u = V/2$ for a series of semicircular vanes.

Assumptions.

1. Steady jet of velocity V and area a; the vane moves at u along the jet.
2. The vane is smooth, so the relative speed of the water over the vane stays constant (equal to its inlet value $V - u$).
3. The water leaves the vane at angle θ to the direction of motion, moving backwards relative to the vane. A semicircular vane has $\theta = 0$.

Step 1. Single curved plate: relative inlet velocity. Seen from the vane, the water enters along x with speed $(V-u)$. The mass reaching the vane is $\rho a(V-u)$ per second (as for the single flat plate).

Step 2. Relative outlet velocity. The relative velocity turns through the vane, keeping its size $(V-u)$. Its component along x is backwards:

$$V_{r2,x} = -(V-u)\cos\theta$$

Step 3. Absolute outlet velocity along x. Add the vane velocity u to the relative velocity:

$$V_{2x} = u-(V-u)\cos\theta$$

Step 4. Force. The inlet absolute velocity along x is V. So

$$F = \rho a(V-u)[V-u+(V-u)\cos\theta] = \rho a(V-u)^2(1+\cos\theta)$$

Step 5. Series of vanes: use the whole jet. The mass is now ρaV, and the change of velocity along x is the same as in Step 4:

$$F = \rho aV[(V-u)+(V-u)\cos\theta] = \rho aV(V-u)(1+\cos\theta)$$

Step 6. Semicircular vanes: $\theta = 0$. With $\cos\theta = 1$:

$$F = 2\rho aV(V-u), \qquad \text{Work/s} = Fu = 2\rho aV(V-u)u$$

Step 7. Efficiency. Divide by the jet kinetic energy rate $\frac{1}{2}\rho aV^3$:

$$\eta = \frac{2\rho aV(V-u)u}{\frac{1}{2}\rho aV^3} = \frac{4u(V-u)}{V^2}$$

Step 8. Maximise. Differentiate $\eta = \frac{4}{V^2}\left(Vu-u^2\right)$:

$$\frac{d\eta}{du} = \frac{4}{V^2}(V-2u) = 0 \quad \Rightarrow \quad u = \frac{V}{2}, \qquad \frac{d^2\eta}{du^2} = -\frac{8}{V^2} < 0$$

Step 9. Maximum value.

$$\eta_{\max} = \frac{4(V/2)(V/2)}{V^2} = 1 = 100\%$$

Result.

$$F = 2\rho aV(V-u), \qquad \eta = \frac{4u(V-u)}{V^2}, \qquad \eta_{\max} = 100\% \text{ at } u = \frac{V}{2}$$

What it tells us. At $u = V/2$ the absolute exit velocity along x is $2u-V = 0$, so the water leaves at rest and gives up all its kinetic energy. This is the ideal limit; real buckets reach about 98% (see Section 3.3).

1.3 Jet on a moving unsymmetrical curved vane (velocity triangles) CORE

IN SIMPLE WORDS

Real turbine vanes are not symmetrical, and the jet often strikes them at an angle. To get the force we need the velocity of the water leaving the vane. The velocity triangles give it. At inlet the water must slide onto the vane smoothly, with no shock. At outlet it slides off along the vane tip. The force then follows from the change in the whirl (forward) component of the velocity, and the efficiency from the work divided by the jet energy.

The jet enters at angle α to the direction of vane motion. For shock-free entry and exit, the relative velocity must be tangential to the vane tip at inlet (angle θ) and outlet (angle ϕ).

- Inlet triangle: $V_{w1} = V_1 \cos\alpha$, $V_{f1} = V_1 \sin\alpha$, $\tan\theta = V_{f1}/\left(V_{w1} - u_1\right)$
- Outlet triangle: $V_{r2} = V_{r1}$ (smooth vane), $V_{w2} = V_{r2}\cos\phi - u_2$
- $u_1 = u_2 = u$ for a vane moving in a straight line

$$F_x = \rho a V_1 \left(V_{w1} \pm V_{w2}\right)$$

$$\text{Work/s} = \rho a V_1 \left(V_{w1} \pm V_{w2}\right) u$$

$$\eta = \frac{2\left(V_{w1} \pm V_{w2}\right) u}{V_1^2}$$

Take $+$ when V_{w2} is opposite to the vane motion u, $-$ when V_{w2} is in the same direction as u, and $V_{w2} = 0$ when the jet leaves at 90° to the motion.

Derivation from scratch: force, work and efficiency for a jet on a moving unsymmetrical curved vane

Aim. Prove $F_x = \rho a V_1\left(V_{w1} \pm V_{w2}\right)$, $\text{Work/s} = F_x u$ and $\eta = 2\left(V_{w1} \pm V_{w2}\right) u/V_1^2$.

Assumptions.

1. A series of identical vanes on a wheel, so the whole jet of mass flow $\rho a V_1$ is used.
2. The vane is smooth: the relative speed is unchanged from inlet to outlet, $V_{r2} = V_{r1}$.
3. Shock-free entry: $\vec{V}_{r1}$ is tangent to the vane at the inlet tip. The water leaves along the outlet tip at angle ϕ to the direction of motion.
4. $u_1 = u_2 = u$, and the x-axis is the direction of vane motion.

Step 1. Inlet triangle. The jet speed V_1 makes angle α with x:

$$V_{w1} = V_1\cos\alpha, \qquad V_{f1} = V_1\sin\alpha$$

Step 2. Relative velocity at inlet. Subtract the vane velocity u from the whirl component. The relative velocity has components $\left(V_{w1} - u\right)$ along x and V_{f1} across it, so

$$\tan\theta = \frac{V_{f1}}{V_{w1} - u}, \qquad V_{r1} = \sqrt{\left(V_{w1} - u\right)^2 + V_{f1}^2}$$

Step 3. Outlet relative velocity. The vane is smooth, so $V_{r2} = V_{r1}$. At the outlet tip the relative velocity points backwards at angle ϕ to the motion, so its x-component is $-V_{r2}\cos\phi$.

Step 4. Absolute whirl at outlet. Add the vane velocity u to the relative velocity along x:

$$V_{2x} = u - V_{r2}\cos\phi$$

Define $V_{w2} = V_{r2}\cos\phi - u$. When this is positive, the water leaves backwards (opposite to u), and $V_{2x} = -V_{w2}$.

Step 5. Momentum equation. The inlet velocity along x is V_{w1}, and the outlet velocity along x is $-V_{w2}$:

$$F_x = \rho a V_1 \left[V_{w1} - \left(-V_{w2} \right) \right] = \rho a V_1 \left(V_{w1} + V_{w2} \right)$$

If the water leaves forwards (same direction as u) its magnitude V_{w2} is subtracted, which gives the $-$ sign. If $V_{w2} = 0$, only V_{w1} remains.

Step 6. Work per second. The force moves its point of application u metres every second:

$$\text{Work/s} = F_x\, u = \rho a V_1 \left(V_{w1} \pm V_{w2} \right) u$$

Step 7. Efficiency. The jet kinetic energy supplied per second is $\tfrac{1}{2}\rho a V_1 V_1^2$. Therefore

$$\eta = \frac{\rho a V_1 \left(V_{w1} \pm V_{w2} \right) u}{\tfrac{1}{2}\rho a V_1^3} = \frac{2\left(V_{w1} \pm V_{w2} \right) u}{V_1^2}$$

Result.

$$F_x = \rho a V_1 \left(V_{w1} \pm V_{w2} \right), \quad \text{Work/s} = \rho a V_1 \left(V_{w1} \pm V_{w2} \right) u$$

$$\eta = \frac{2\left(V_{w1} \pm V_{w2} \right) u}{V_1^2}$$

What it tells us. The force depends on how much the vane reverses the whirl component. The ideal vane would take V_{w2} as large and opposite as possible. For the best efficiency, the water should leave with as little absolute velocity as possible.

Worked examples

Step-by-step method.

1. Convert all data to SI: mm to m, L/s to m^3/s, rpm to rad/s.
2. Find the jet area $a = \pi d^2/4$ and the jet speed V.
3. Decide the case: fixed or moving, flat or curved, single plate or series of plates.
4. Write the relative velocity $(V - u)$ if the vane moves, and draw the inlet and outlet velocity triangles for a curved vane.
5. Apply $F = \dot{m}\left(V_{\text{in}} - V_{\text{out}} \right)$ along the jet direction.
6. Work per second $= F \times u$; efficiency $=$ work per second $/ \tfrac{1}{2}\rho a V^3$.
7. Check units and that $\eta \le 100\%$.

Example 1.0 (warm-up) CORE A $50\ \text{mm}$ jet at $20\ \text{m/s}$ strikes a fixed flat plate normally. Find the force.

1. $a = \dfrac{\pi}{4}\,(\,0.05\,)^{\,2} = 1.9635 \times 10^{-3}\ \text{m}^2$
2. $\dot{m} = \rho a V = 1000 \times 1.9635 \times 10^{-3} \times 20 = 39.27\ \text{kg/s}$
3. $F = \dot{m} V = 39.27 \times 20 = \ 785.4\ \text{N}$

Example 1.1 CORE A 75 mm jet with velocity 20 m/s strikes a fixed flat plate normally. Find the force. If the plates are a series moving at 8 m/s, find force, power and efficiency.

1. $a = \dfrac{\pi}{4}\,(\,0.075\,)^{\,2} = 4.418 \times 10^{-3}\ \text{m}^2$
2. Fixed plate: $F = \rho a V^2 = 1000 \times 4.418 \times 10^{-3} \times 20^2 = \ 1767\ \text{N}$
3. Series of moving plates: $F = \rho a V\,(\,V - u\,) \ = 1000 \times 4.418 \times 10^{-3} \times 20 \times 12 = \ 1060\ \text{N}$
4. Power $= F \times u = 1060 \times 8 = \ 8.48\ \text{kW}$; $\eta = \dfrac{2u\,(\,V-u\,)}{V^2} = \dfrac{2 \times 8 \times 12}{400} = \ 48\%$

Example 1.2 CORE A jet at 40 m/s strikes a curved vane moving at 20 m/s. The jet makes 30° with the vane motion at inlet and leaves at 90° to the motion at outlet. Find the vane angles for shock-free flow and the work done per unit weight.

1. $V_{w1} = 40\cos 30^\circ = 34.64\ \text{m/s}$; $V_{f1} = 40 \sin 30^\circ = 20\ \text{m/s}$
2. $\tan\theta = \dfrac{20}{34.64 - 20} = 1.366$, so $\theta = 53.8^\circ$
3. $V_{r1} = \sqrt{14.64^2 + 20^2} = 24.79\ \text{m/s} = V_{r2}$
4. Exit at 90° means $V_{w2} = 0$, so $\cos\phi = u/V_{r2} = 20/24.79$, giving $\phi = 36.2^\circ$
5. Work per unit weight $= \dfrac{V_{w1}\,u}{g} = \dfrac{34.64 \times 20}{9.81} = \ 70.6\ \text{N}\cdot\text{m/N}$

Common mistakes and exam tips.

- Using V instead of $V - u$ for a moving plate. A single moving plate gets $\rho a\,(\,V-u\,)^{\,2}$; a series of plates uses the whole jet, $\rho a V\,(\,V-u\,)$.
- Forgetting to change mm to m, or using $a = \pi d^2$ instead of $a = \pi d^2/4$.
- Taking the wrong sign for V_{w2}. Check whether the water leaves against the vane motion (+) or along it (−).
- Measuring an angle from the wrong line. Here θ and ϕ are measured from the direction of vane motion, not from the vertical.
- Dividing the work by $\rho a V^3$ instead of $\frac{1}{2}\rho a V^3$ when finding efficiency.

Check your understanding.

1. CORE What happens to the force on a fixed flat plate if the jet speed doubles? *(It becomes four times, since $F \propto V^2$.)*
2. CORE Why is the force on a fixed semicircular plate twice that on a flat plate? *(The jet is turned back through 180°, so its velocity changes by $2V$, not V.)*
3. HOTS A series of semicircular vanes moves at $u = V/2$. What is the absolute exit velocity along the jet, and what does it mean? *(It is $2u - V = 0$: the water leaves at rest, so $\eta = 100\%$.)*

2. Hydraulic turbines: classification, heads and efficiencies (L3) CORE

A hydraulic turbine converts the hydraulic energy of water into mechanical energy at the shaft. Turbines are grouped four ways, and each syllabus turbine has a fixed place in every scheme.

> **IN SIMPLE WORDS**
> A turbine is a water-driven wheel that turns a generator. High, narrow flows from tall dams suit Pelton wheels. Medium heads suit Francis turbines. Low heads with huge flows (rivers) suit Kaplan turbines. We sort turbines by how the water gives up its energy (impulse or reaction), by flow direction, by head and by specific speed. Efficiency tells us how much of the water's power finally reaches the shaft.

Basis	Types	Example
Energy at runner inlet	Impulse (all kinetic; runner at atmospheric pressure) · Reaction (kinetic + pressure; runner full of water)	Pelton · Francis, Kaplan
Direction of flow through runner	Tangential · Radial (inward/outward) · Axial · Mixed	Pelton · old Francis · Kaplan · modern Francis
Head at inlet	High (above 250 m) · Medium (60–250 m) · Low (below 60 m)	Pelton · Francis · Kaplan
Specific speed (SI, kW)	Low (8.5–51) · Medium (51–255) · High (255–860)	Pelton · Francis · Kaplan

Impulse vs reaction: in an impulse turbine the entire available head is converted to kinetic energy in a nozzle, and the pressure stays atmospheric across the runner. In a reaction turbine only part of the head is converted before the runner; the rest is converted while water flows through the runner, so the runner must be enclosed and run full.

2.1 Heads CORE

> **IN SIMPLE WORDS**
> Gross head is the full height drop from the reservoir to the tailrace. Water rubs on the pipe wall while flowing down the penstock, so some head is lost. What is left is the net head, and this is the head that the turbine really works with. All turbine formulas use the net head H.

- Gross head H_g: difference between headrace and tailrace levels
- Net (effective) head: $H = H_g - h_f$, where $h_f = fLV^2/(2gD)$ is the penstock friction loss (Bansal writes $4fLV^2/(2gD)$ with f as the friction coefficient)

2.2 Efficiencies CORE

IN SIMPLE WORDS

Energy is lost at several places on the way from water to shaft. We give each loss its own efficiency. Hydraulic efficiency counts losses in the water passages. Mechanical efficiency counts bearing friction. Volumetric efficiency counts water that leaks past the runner. Overall efficiency is the product of the efficiencies: it compares the shaft power with the power of the water.

$$\text{W.P.} = \frac{\rho g Q H}{1000}\,\text{kW}$$

Derivation from scratch: power available in the water (water power)

Aim. Prove $\text{W.P.} = \rho g Q H/1000$ kW.

Assumptions.

1. Steady discharge Q (m^3/s) under a constant net head H (m).
2. Water density $\rho = 1000\,\text{kg/m}^3$.

Step 1. Mass per second. Volume per second times density:

$$\dot{m} = \rho Q \quad (\,\text{kg/s}\,)$$

Step 2. Weight per second. Weight is mass times g:

$$\text{weight/s} = \rho g Q \quad (\,\text{N/s}\,)$$

Step 3. Energy per newton. Each newton of water at head H can do H newton-metres of work. This is the meaning of "head": energy per unit weight.

Step 4. Power in watts. Energy per second equals (weight per second) $\times$ (energy per newton):

$$\text{W.P.} = \rho g Q \times H \quad (\,\text{N/s} \times \text{m} = \text{W}\,)$$

Step 5. Convert to kW. Divide by 1000.

Result.

$$\text{W.P.} = \frac{\rho g Q H}{1000}\,\text{kW}$$

What it tells us. Water power grows in direct proportion to both discharge and head. Doubling either one doubles the power available.

Efficiency	Definition	Expression
Hydraulic η_h	Runner power / water power	$\rho Q\,(\,V_{w1}u_1 \pm V_{w2}u_2\,)\,/\,(\,\rho g Q H\,)$
Mechanical η_m	Shaft power / runner power	S.P./R.P.
Volumetric η_v	Water actually striking runner / water supplied	$(\,Q-\Delta Q\,)\,/Q$
Overall η_o	Shaft power / water power	$\eta_h \times \eta_m$ ($\times\ \eta_v$ if leakage is counted)

2.3 Euler's turbine equation CORE

IN SIMPLE WORDS

The water turns the runner because it changes its swirl (whirl) as it passes through. Fast swirl at the inlet and little swirl at the outlet means the runner took the angular momentum. The torque equals the angular momentum given up per second. Multiply by the angular speed ω to get power. Here V_w is the swirl speed and u is the blade speed at the same radius.

Applying the angular-momentum equation to the runner, $\text{torque} = \rho Q\,(\,V_{w1}r_1 \pm V_{w2}r_2\,)$. Multiplying by ω gives the work done per second, and dividing by $\rho g Q$ gives work per unit weight:

$$\frac{\text{Work}}{\text{unit weight}} = \frac{1}{g}\,(\,V_{w1}u_1 \pm V_{w2}u_2\,), \qquad \eta_h = \frac{V_{w1}u_1 \pm V_{w2}u_2}{gH}$$

Derivation from scratch: Euler's turbine equation

Aim. Prove that the work done per unit weight of water is $(\,V_{w1}u_1 \pm V_{w2}u_2\,)\,/g$ and that $\eta_h = (\,V_{w1}u_1 \pm V_{w2}u_2\,)\,/\,(\,gH\,)$.

Assumptions.

1. Steady flow through the runner, discharge Q, water enters at radius r_1 and leaves at radius r_2.
2. Only the whirl (tangential) velocity components V_{w1}, V_{w2} have a moment arm about the shaft. The flow components V_f point towards or along the axis, so they exert no torque.
3. The runner turns at constant angular speed ω, so $u_1 = \omega r_1$ and $u_2 = \omega r_2$.

Step 1. Angular momentum entering per second. Mass per second ρQ carries moment of momentum $V_{w1}r_1$ per kg:

$$\text{in} = \rho Q\, V_{w1}\, r_1$$

Step 2. Angular momentum leaving per second. Take whirl as positive in the direction of rotation:

$$\text{out} = \rho Q\, V_{w2}\, r_2$$

Step 3. Torque. Torque on the water equals the rate of change of angular momentum, out minus in. The torque on the runner is equal and opposite:

$$T = \rho Q \left(V_{w1} r_1 - V_{w2} r_2 \right)$$

If the outlet whirl is opposite to the rotation, its magnitude is added instead. We write both cases as $T = \rho Q \left(V_{w1} r_1 \pm V_{w2} r_2 \right)$: the $+$ sign is for outlet whirl opposite to blade motion and the $-$ sign for outlet whirl in the direction of blade motion.

Step 4. Power. Power is torque times angular speed. Use $u = \omega r$:

$$P = T\omega = \rho Q \left(V_{w1}\, \omega r_1 \pm V_{w2}\, \omega r_2 \right) = \rho Q \left(V_{w1} u_1 \pm V_{w2} u_2 \right)$$

Step 5. Work per unit weight. Divide the power by the weight of water per second, $\rho g Q$. The ρQ cancels:

$$\frac{P}{\rho g Q} = \frac{1}{g} \left(V_{w1} u_1 \pm V_{w2} u_2 \right)$$

Step 6. Hydraulic efficiency. Runner power divided by water power $\rho g Q H$:

$$\eta_h = \frac{\rho Q \ \left(\ V_{w1} u_1 \pm V_{w2} u_2 \ \right)}{\rho g Q H} = \frac{V_{w1} u_1 \pm V_{w2} u_2}{g H}$$

Step 7. Radial or axial discharge. If the water leaves with no whirl, $V_{w2} = 0$, and the second term vanishes.

Result.

$$\frac{\text{Work}}{\text{unit weight}} = \frac{1}{g} \ \left(\ V_{w1} u_1 \pm V_{w2} u_2 \ \right), \qquad \eta_h = \frac{V_{w1} u_1 \pm V_{w2} u_2}{g H}$$

What it tells us. The work depends only on the whirl components and blade speeds at inlet and outlet, not on the shape of the passage in between. Taking $V_{w2} = 0$ gives $V_{w1} u_1 / g$, and the water leaves with its speed all in the flow direction, so no energy is wasted as swirl.

- Plus ($+$) sign: outlet whirl opposite to blade motion (Pelton bucket)
- Minus ($-$) sign: outlet whirl in the direction of blade motion
- Radial or axial discharge ($V_{w2} = 0$): $\text{work} = V_{w1} u_1 / g$ — the design condition for Francis and Kaplan turbines, because it gives minimum exit loss

Common mistakes and exam tips.

- Using the gross head H_g in $\eta_h = \dfrac{V_{w1}u_1}{gH}$ and in water power. Use the net head H.
- Mixing up the three efficiencies. Remember $\eta_o = \eta_h \times \eta_m$ and that shaft power $= \eta_o \times$ water power.
- Forgetting to divide by 1000 to get kW from $\rho g Q H$.
- Using the wrong sign in Euler's equation. Plus when the outlet whirl opposes the blade motion, minus when it agrees with it.
- Saying that the pressure is atmospheric on a Francis runner. Only an impulse runner works at atmospheric pressure.

Check your understanding.

1. CORE In which type of turbine is the runner at atmospheric pressure? *(Impulse, such as a Pelton wheel.)*
2. CORE If $\eta_h = 0.90$ and $\eta_m = 0.95$, what is η_o? *($0.90 \times 0.95 = 0.855$, i.e. 85.5%.)*
3. HOTS Why do Francis and Kaplan designers aim at $V_{w2} = 0$? *(The water then carries no swirl energy out, so exit loss is minimum and the work is $V_{w1}u_1/g$.)*

3. Pelton wheel — impulse turbine (L4–L5) CORE

The Pelton wheel is a tangential-flow impulse turbine for high heads (above about 250 m) and low discharge. Its maximum hydraulic efficiency occurs when the bucket speed is half the jet speed.

IN SIMPLE WORDS

A Pelton wheel is a wheel with cups on its rim. A fast water jet from a nozzle hits the cups and pushes the wheel round. All the pressure energy is changed to speed in the nozzle first, so the wheel runs in air. Think of a water mill, but with a jet of high pressure instead of a stream. The cups turn the water almost fully back, which gives the biggest push. The best wheel speed is about half the jet speed.

3.1 Constructional details CORE

IN SIMPLE WORDS

Each part has one job. The nozzle makes the jet. The needle (spear) opens or closes the nozzle to control flow. The buckets catch the jet. The splitter divides the jet into two halves so the wheel is not pushed sideways. The casing stops splashes. The brake jet stops the wheel quickly. The deflector protects the penstock when load suddenly drops.

1. **Nozzle and spear** — converts the full head into a high-velocity jet. The spear (needle) moves axially to vary the jet area and hence discharge.

2. **Runner with buckets** — a circular disc with double-hemispherical (cup) buckets on its rim. A central **splitter** divides the jet into two equal halves, so axial thrust cancels. Each half is deflected through 160–165° (not 180°, so the leaving water does not hit the next bucket).
3. **Casing** — prevents splashing and guides water to the tailrace. It has no hydraulic function, since the runner is at atmospheric pressure.
4. **Breaking jet** — a small nozzle that strikes the back of the buckets to bring the runner to rest quickly after the main jet is shut.
5. **Deflector** — swings into the jet on sudden load rejection (part of governing).

3.2 Velocity triangles CORE

> **IN SIMPLE WORDS**
>
> The velocity triangle at the inlet is just a straight line, because the jet moves along the same line as the bucket. At the outlet the bucket turns the water almost back, so the triangle is narrow. The speed of the water relative to the bucket, V_r, is the jet speed minus the bucket speed. A rough bucket slows the water slightly, which is the factor k.

Inlet (jet and bucket motion are in the same line, so the triangle is a straight line):

$$V_1 = C_v\sqrt{2gH}, \quad u = \frac{\pi DN}{60}, \quad V_{w1} = V_1, \quad V_{r1} = V_1 - u, \quad \alpha = 0,\ \theta = 0$$

C_v = coefficient of velocity of nozzle, 0.97–0.99. D = pitch circle diameter of the wheel.

Outlet ($u_2 = u_1 = u$, since inlet and outlet are at the same radius):

$$V_{r2} = k\,V_{r1}, \qquad V_{w2} = V_{r2}\cos\phi - u$$

ϕ = bucket outlet angle ($180^\circ -$ deflection angle), k = bucket friction factor ($k = 1$ for smooth bucket).

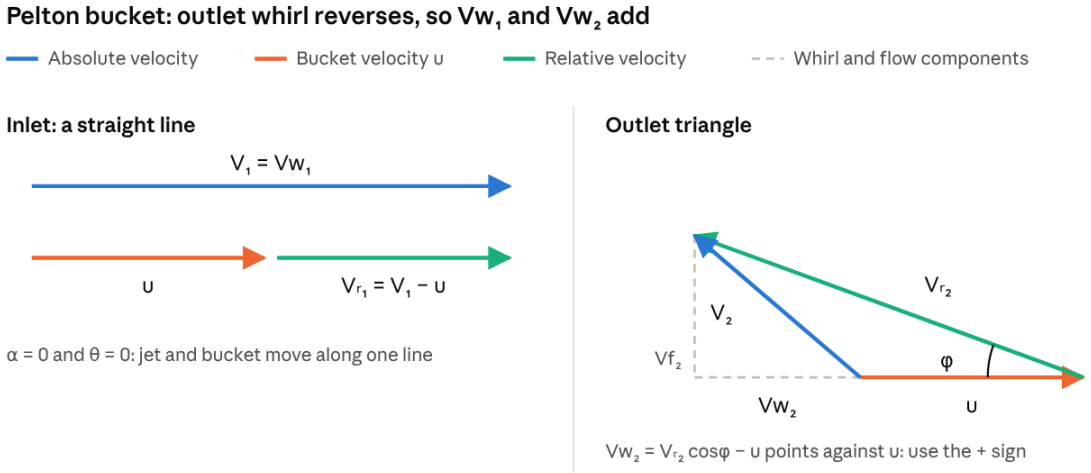


Figure 1. Pelton wheel velocity diagrams at inlet and outlet

At inlet all velocities lie on one line; at outlet the bucket turns the relative velocity back through 160–165°, so the whirl component reverses and adds to V_{w1} (diagram not to scale).

Derivation from scratch: Pelton jet velocity and velocity triangles

Aim. Prove $V_1 = C_v\sqrt{2gH}$, $u = \pi DN/60$, $V_{r1} = V_1 - u$ and $V_{w2} = V_{r2}\cos\phi - u$.

Assumptions.

1. The net head H at the nozzle is converted into the jet in the nozzle; the jet leaves at atmospheric pressure at nearly the same level.
2. Nozzle friction is included only through the coefficient C_v.
3. The jet is tangential to the pitch circle, so the jet and the bucket move along one line at the inlet.
4. Inlet and outlet are at the same radius, so $u_1 = u_2 = u$.

Step 1. Energy equation at the nozzle. Take the supply point (pressure head plus velocity head equal to the net head H) and the jet outlet (atmospheric pressure, gauge pressure zero). Without losses,

$$H = \frac{V_{\text{th}}^2}{2g} \quad \Rightarrow \quad V_{\text{th}} = \sqrt{2gH}$$

Step 2. Include nozzle friction. The actual speed is smaller. The ratio is the coefficient of velocity $C_v = V_1/V_{\text{th}}$, so

$$V_1 = C_v\sqrt{2gH}$$

Step 3. Blade speed. The angular speed is $\omega = 2\pi N/60$ rad/s. The speed of a point at the pitch circle (radius $D/2$) is

$$u = \omega\frac{D}{2} = \frac{2\pi N}{60}\times\frac{D}{2} = \frac{\pi DN}{60}$$

Step 4. Inlet triangle. The jet is along the bucket motion, so the whole jet velocity is whirl velocity, and the flow component is zero:

$$V_{w1} = V_1, \qquad V_{f1} = 0, \qquad \alpha = 0,\ \theta = 0$$

The relative velocity is the difference of the two collinear velocities:

$$V_{r1} = V_1 - u$$

Step 5. Outlet relative velocity. The bucket friction reduces the relative speed slightly. We write this with a factor $k \le 1$:

$$V_{r2} = k\,V_{r1}$$

Step 6. Outlet whirl. The relative velocity leaves the bucket at angle ϕ to the direction of motion, pointing backwards. Its backward component is $V_{r2}\cos\phi$. The bucket itself moves forward at u. So the net backward speed of the water is

$$V_{w2} = V_{r2}\cos\phi - u$$

(positive when the water leaves against the direction of u).

Result.

$$V_1 = C_v\sqrt{2gH}, \quad u = \frac{\pi DN}{60}, \quad V_{r1} = V_1 - u, \quad V_{r2} = k\,V_{r1}, \quad V_{w2} = V_{r2}\cos\phi - u$$

What it tells us. The jet speed depends only on the head, not on the discharge. The bucket speed depends on the wheel diameter and rpm. Water leaving backwards ($V_{w2} > 0$) means the bucket has turned the jet by more than a flat plate would.

3.3 Work, power and efficiency CORE

IN SIMPLE WORDS

The push on the bucket comes from the total change of whirl speed: the jet comes in at V_1 and leaves going backwards at V_{w2}. The power is this push times the bucket speed u. Efficiency compares this power with the energy of the jet. If the bucket stands still, the power is zero. If it runs as fast as the jet, there is no push, so power is also zero. The best speed is between: $u = V_1/2$.

V_{w2} acts opposite to u, so the $+$ sign of the Euler equation applies.

$$F_x = \rho a V_1 \left(V_{w1} + V_{w2} \right), \qquad P = \rho a V_1 \left(V_{w1} + V_{w2} \right) u$$

Substituting $V_{w1} + V_{w2} = \left(V_1 - u \right) \left(1 + k \cos \phi \right)$:

$$\eta_h = \frac{2 \left(V_{w1} + V_{w2} \right) u}{V_1^2} = \frac{2 \left(V_1 - u \right) \left(1 + k \cos \phi \right) u}{V_1^2}$$

Condition for maximum efficiency: set $d\eta_h/du = 0$, which gives $V_1 - 2u = 0$.

$$u = \frac{V_1}{2}, \qquad \eta_{h,\,\max} = \frac{1 + k \cos \phi}{2}$$

With $k = 1$ and $\phi = 15^\circ$, $\eta_{h,\,\max} = 98.3\%$. In practice the best speed ratio is slightly lower ($u \approx 0.46\, V_1$) because of bucket friction and windage.

Derivation from scratch: Pelton work done, hydraulic efficiency and condition for maximum efficiency

Aim. Prove $\eta_h = 2 \left(V_1 - u \right) \left(1 + k \cos \phi \right) u / V_1^2$, then that η_h is maximum at $u = V_1/2$ with $\eta_{h,\,\max} = \left(1 + k \cos \phi \right) / 2$.

Assumptions.

1. Steady jet of velocity V_1, discharge $Q = a V_1$, hitting buckets that follow each other so that the whole jet is used.
2. The outlet whirl V_{w2} is opposite to u, so the $+$ sign applies.
3. Velocity triangles as in Section 3.2: $V_{w1} = V_1$, $V_{r1} = V_1 - u$, $V_{r2} = k V_{r1}$, $u_1 = u_2 = u$.
4. The energy supplied by the jet is its kinetic energy $\frac{1}{2} \dot{m} V_1^2$ (nozzle loss is already inside C_v).

Step 1. Total whirl change. Add the inlet whirl and the (reversed) outlet whirl:

$$V_{w1} + V_{w2} = V_1 + (V_{r2} \cos \phi - u) = V_1 + k V_{r1} \cos \phi - u$$

Step 2. Insert $V_{r1} = V_1 - u$.

$$V_{w1} + V_{w2} = \left(V_1 - u \right) + k \left(V_1 - u \right) \cos\phi = \left(V_1 - u \right) \left(1 + k \cos\phi \right)$$

Step 3. Force on the buckets. Mass per second is $\dot{m} = \rho a V_1 = \rho Q$. Impulse-momentum along the motion:

$$F_x = \rho a V_1 \left(V_{w1} + V_{w2} \right)$$

Step 4. Work done per second (power). Force times bucket speed:

$$P = F_x \, u = \rho a V_1 \left(V_{w1} + V_{w2} \right) u = \rho Q \left(V_{w1} + V_{w2} \right) u$$

Per kg/s of water, the work done is $\left(V_{w1} + V_{w2} \right) u$ joules per kg.

Step 5. Hydraulic efficiency. Power divided by the jet kinetic energy per second, $\frac{1}{2}\rho a V_1 V_1^2$:

$$\eta_h = \frac{\rho a V_1 \left(V_{w1} + V_{w2} \right) u}{\frac{1}{2}\rho a V_1^3} = \frac{2 \left(V_{w1} + V_{w2} \right) u}{V_1^2} = \frac{2 \left(V_1 - u \right) \left(1 + k \cos\phi \right) u}{V_1^2}$$

Step 6. Prepare to differentiate. For a given wheel k, ϕ and V_1 are fixed. Let $A = 2 \left(1 + k \cos\phi \right) / V_1^2$, a constant:

$$\eta_h = A \left(V_1 u - u^2 \right)$$

Step 7. First derivative equal to zero.

$$\frac{d\eta_h}{du} = A \left(V_1 - 2u \right) = 0 \quad \Rightarrow \quad u = \frac{V_1}{2}$$

Step 8. Second derivative. Because $A > 0$,

$$\frac{d^2\eta_h}{du^2} = -2A < 0$$

so this is a maximum.

Step 9. Maximum value. Put $u = V_1/2$, so $V_1 - u = V_1/2$:

$$\eta_{h,\,\max} = \frac{2 \left(V_1/2 \right) \left(1 + k \cos\phi \right) \left(V_1/2 \right)}{V_1^2} = \frac{1 + k \cos\phi}{2}$$

Result.

$$\eta_h = \frac{2 \left(V_1 - u \right) \left(1 + k \cos\phi \right) u}{V_1^2}, \qquad u = \frac{V_1}{2}, \qquad \eta_{h,\,\max} = \frac{1 + k \cos\phi}{2}$$

What it tells us. Efficiency is highest when the bucket runs at half the jet speed. It grows as ϕ gets smaller (the jet is turned back more) and as k gets closer to 1 (less friction). Checks: $\phi = 90^\circ$ gives 50% (flat plate, Section 1.2), and $\phi = 0$ with $k = 1$ gives 100% (semicircular vane). With $k = 1$ and $\phi = 15^\circ$, $\eta_{h,\,\max} = \left(1 + 0.9659 \right)/2 = 98.3\%$.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

3.4 Design relations ADVANCED

Quantity	Relation / typical value
Speed ratio K_u	$u/\sqrt{2gH} = 0.43$–0.48
Jet ratio m	$D/d = 11$–14 (12 is common)
Number of buckets z	$15 + D/(2d) = 15 + 0.5m$ (Tygun's formula)
Bucket width / depth	about $5d$ / $1.2d$
Discharge (one jet)	$Q = (\pi/4)\, d^2 \times V_1$
Number of jets	Q total / Q per jet; use multiple jets if m falls below about 10

3.5 Governing of the Pelton wheel CORE

IN SIMPLE WORDS

A generator must run at a fixed speed to give steady frequency. When the electrical load falls, the wheel speeds up. The governor senses this and closes the needle a little, so less water flows and the speed comes back. When the load rises, it does the opposite. If the load drops suddenly, a deflector first diverts the jet so the long pipe does not get a pressure shock.

Governing keeps the runner speed constant (synchronous speed of the generator) as load changes, by changing the discharge. Large Pelton wheels use an **oil-pressure governor**:

1. Load on the generator falls, so runner speed rises.
2. The centrifugal governor (flyballs) moves outward, lifting the sleeve.
3. The sleeve lever opens the **control (relay) valve**, sending pressurised oil from the gear pump to one side of the **servomotor** piston.
4. The servomotor piston pushes the spear forward, reducing the nozzle area and discharge.
5. Speed returns to normal; the sleeve and control valve return to neutral.

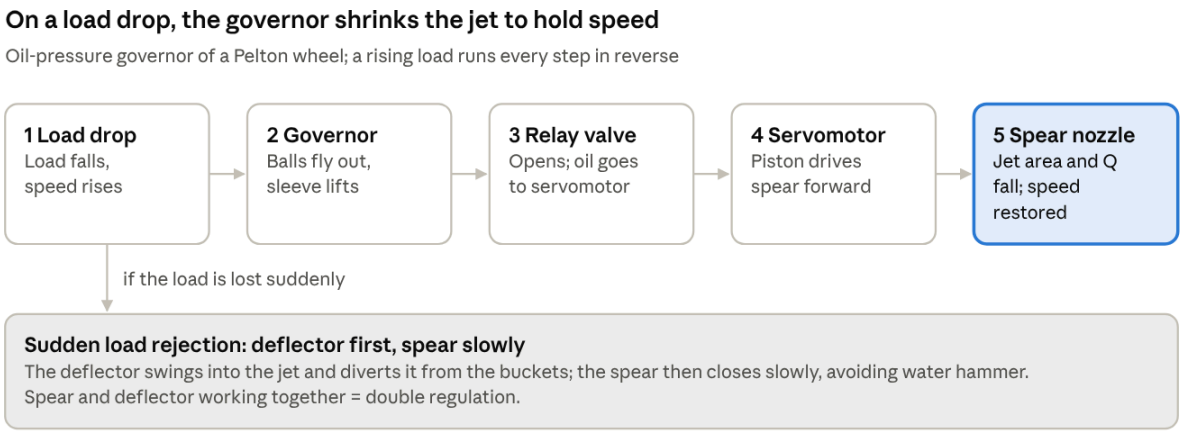


Figure 2. Oil-pressure governing of a Pelton wheel on a load drop

Rising load reverses every step. **Sudden load rejection:** closing the spear quickly would cause water hammer in the long penstock. So the **deflector** first diverts the jet away from the buckets, and the spear then closes slowly. Operating spear and deflector together is called **double regulation**.

Worked examples

Step-by-step method.

1. Convert units: N in rpm, P in kW (or W), Q in $\mathrm{m^3/s}$, diameters in m.
2. Find the jet velocity $V_1 = C_v\sqrt{2gH}$.
3. Find the bucket speed u from $u = \pi DN/60$ or from the speed ratio $K_u\sqrt{2gH}$.
4. Find $V_{r1} = V_1 - u$ and $V_{r2} = kV_{r1}$, with $\phi = 180^\circ - \text{deflection}$.
5. Find $V_{w2} = V_{r2}\cos\phi - u$, then $V_{w1} + V_{w2}$.
6. Power $= \rho Q\,(\,V_{w1} + V_{w2}\,)\,u$ and $\eta_h = 2\,(\,V_{w1} + V_{w2}\,)\,u/V_1^2$.
7. For design problems, use $Q = P/\left(\rho g H \eta_o\right)$ and $Q = (\,\pi/4\,)\,d^2 V_1$ for the jet diameter.

Example 3.0 (warm-up) CORE A Pelton wheel of pitch diameter $1.5\ \mathrm{m}$ runs at $400\ \mathrm{rpm}$. Find the bucket speed, the jet speed for maximum efficiency, and that efficiency if $k = 1$ and $\phi = 15^\circ$.

1. $u = \dfrac{\pi DN}{60} = \dfrac{\pi \times 1.5 \times 400}{60} = 31.42\ \mathrm{m/s}$
2. For maximum efficiency $u = V_1/2$, so $V_1 = 2u = 62.83\ \mathrm{m/s}$
3. $\eta_{h,\ \max} = \dfrac{1 + k\cos\phi}{2} = \dfrac{1 + \cos 15^\circ}{2} = 98.3\%$

Example 3.1 CORE A Pelton wheel has mean bucket speed 35 m/s, jet discharge 150 L/s and head 300 m. Buckets deflect the jet through 160°. Take $C_v = 0.98$. Find the power and hydraulic efficiency.

1. $V_1 = 0.98\sqrt{2 \times 9.81 \times 300} = 75.19\ \mathrm{m/s}$; $u = 35\ \mathrm{m/s}$

2. $V_{r1} = 75.19 - 35 = 40.19\ \text{m/s} = V_{r2}$; $\phi = 180^\circ - 160^\circ = 20^\circ$
3. $V_{w2} = 40.19 \cos 20^\circ - 35 = 2.77\ \text{m/s}$
4. $P = \rho Q\ (\ V_{w1} + V_{w2}\ )\ u = 1000 \times 0.15 \times\ (\ 75.19 + 2.77\ )\ \times 35 =\ 409.3\ \text{kW}$
5. $\eta_h = \dfrac{2\ (\ 77.96\ )\ (\ 35\ )}{75.19^2} =\ 96.5\%$

Example 3.2 (design) HOTS Design a single-jet Pelton wheel for shaft power 5000 kW, net head 300 m, speed 300 rpm, $\eta_o = 0.85$, $C_v = 0.98$, $K_u = 0.46$.

1. $Q = \dfrac{P}{\rho g H \eta_o} = \dfrac{5 \times 10^6}{1000 \times 9.81 \times 300 \times 0.85} = 2.00\ \text{m}^3/\text{s}$
2. $V_1 = 0.98\sqrt{2 \times 9.81 \times 300}\ = 75.19\ \text{m/s}$; $d = \sqrt{\dfrac{Q}{0.7854 \times V_1}}\ =\ 0.184\ \text{m}$
3. $u = K_u\sqrt{2gH} = 0.46 \times 76.72 = 35.29\ \text{m/s}$; $D = \dfrac{60u}{\pi N} =\ 2.25\ \text{m}$
4. $m = D/d = 12.2$ (acceptable); $z = 15 + 12.2/2 = 21.1$, so $\ 22\ $ buckets
5. Check: $N_s = \dfrac{N\sqrt{P}}{H^{5/4}} = \dfrac{300\sqrt{5000}}{300^{1.25}} = 17.0$, within the single-jet Pelton range

Common mistakes and exam tips.

- Taking the deflection angle as ϕ. The outlet angle is $\phi = 180^\circ - \text{deflection}$, so a 165° turn gives $\phi = 15^\circ$.
- Writing $V_{r1} = V_1 + u$. The bucket runs away from the jet, so $V_{r1} = V_1 - u$.
- Forgetting C_v in $V_1 = C_v\sqrt{2gH}$, or putting it into u as well.
- Using the wheel (bucket tip) diameter instead of the pitch circle diameter in $u = \pi DN/60$.
- Mixing watts and kilowatts. $\rho Q\ (\ \ldots\ )\ u$ is in W; divide by 1000 for kW.
- Quoting $\eta_h = 100\%$ or $u = V_1/2$ as the practical optimum. In practice $K_u \approx 0.43$–0.48 and η_h is about 0.9 to 0.95.

Check your understanding.

1. CORE A Pelton wheel of $D = 1\ \text{m}$ runs at $600\ \text{rpm}$. What is u? *($u = \pi \times 1 \times 600/60 = 31.42\ \text{m/s}$.)*
2. CORE The buckets turn the jet through 165°. What is ϕ? *($\phi = 180^\circ - 165^\circ = 15^\circ$.)*
3. HOTS For $k = 1$ and $\phi = 0^\circ$, what is $\eta_{h,\ \max}$, and why is it not reached in practice? *($(\ 1 + 1\ )\ /2 = 100\%$. The jet cannot be turned a full 180° because it would hit the next bucket, and there is friction.)*

4. Francis turbine — inward mixed-flow reaction turbine (L6)

CORE

The Francis turbine is a reaction turbine for medium heads (about 60–250 m). Water enters the runner radially inward and leaves axially, and the runner is designed for radial (whirl-free) discharge so that $\text{work} = V_{w1}u_1/g$.

IN SIMPLE WORDS

A Francis turbine is like a closed, water-filled wheel with curved blades. Water goes round the wheel in a spiral casing, passes through adjustable guide vanes that give it a spin, and then flows inward through the runner. On the way it loses both speed and pressure, and the blades take the energy. The water leaves down a draft tube. The speed of the runner at the inlet is higher than at the outlet because the inlet diameter is larger.

4.1 Constructional details CORE

IN SIMPLE WORDS

Water travels in this order: spiral casing, stay vanes, guide vanes, runner, draft tube. The casing spreads water round the runner evenly. The stay vanes hold the casing stiff. The guide vanes set the angle of entry and control flow. The runner extracts the energy. The draft tube leads the water away.

1. **Spiral (scroll) casing** — a closed passage whose area decreases along the flow, so water reaches every guide vane at nearly the same velocity.
2. **Stay vanes** — fixed vanes that direct flow to the guide vanes and carry structural load of the casing.
3. **Guide vanes (wicket gates)** — aerofoil vanes pivoted between two rings. They set the inlet angle α and regulate discharge; they are rotated by the governor through a regulating ring.
4. **Runner** — 16–24 fixed, doubly-curved vanes between a crown and a band, keyed to the shaft. Water enters at outer diameter D_1 and leaves at inner diameter D_2.
5. **Draft tube** — a gradually diverging pipe from runner exit to below tailrace level (see 4.4).

4.2 Velocity triangles CORE

IN SIMPLE WORDS

The guide vanes throw the water at the runner at angle α. Part of this speed is along the rim (V_{w1}) and part is inward (V_{f1}). The runner rim moves at u_1. The water relative to the blade is what is left after subtracting u_1. The blade is shaped to match that direction, so the water enters smoothly. At the outlet the water should have no whirl, so it leaves straight along the axis.

Inlet (at D_1):

$$u_1 = \frac{\pi D_1 N}{60}, \quad V_{w1} = V_1 \cos\alpha, \quad V_{f1} = V_1 \sin\alpha, \quad \tan\theta = \frac{V_{f1}}{V_{w1} - u_1}$$

$V_{w1} > u_1$ gives $\theta < 90^\circ$ (slow runner); $V_{w1} = u_1$ gives $\theta = 90^\circ$ (medium runner); $V_{w1} < u_1$ gives $\theta > 90^\circ$ (fast runner), and then $\tan(180^\circ - \theta) = V_{f1}/(u_1 - V_{w1})$.

Outlet (at D_2, radial discharge):

$$u_2 = \frac{\pi D_2 N}{60}, \quad V_{w2} = 0, \quad V_2 = V_{f2}, \quad \tan\phi = \frac{V_{f2}}{u_2}$$

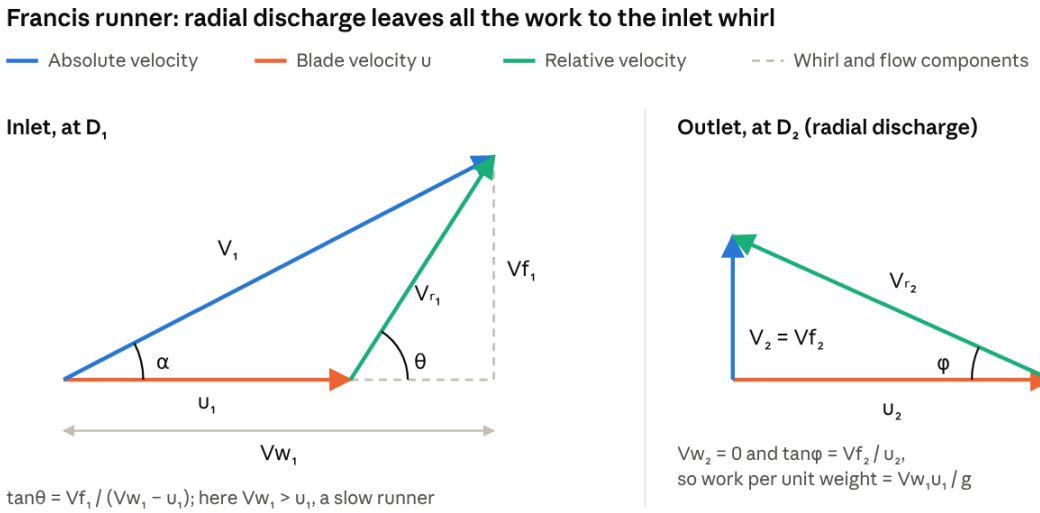


Figure 3. Francis turbine velocity triangles at inlet and outlet

Guide vanes set α at inlet; the runner vane angles θ and ϕ are chosen so water enters without shock and leaves with no whirl (diagram not to scale). A Kaplan turbine has the same triangles with $u_1 = u_2$.

Derivation from scratch: Francis inlet and outlet velocity-triangle relations

Aim. Prove $V_{w1} = V_1 \cos\alpha$, $V_{f1} = V_1 \sin\alpha$, $\tan\theta = V_{f1}/(V_{w1} - u_1)$ and $\tan\phi = V_{f2}/u_2$.

Assumptions.

1. Steady flow, water enters with no shock: the relative velocity is tangent to the vane at the inlet tip and at the outlet tip.
2. The angle α of the guide vane is measured from the direction of the rim velocity u_1.
3. Radial discharge at the outlet: $V_{w2} = 0$.

Step 1. Resolve the inlet velocity. The absolute velocity V_1 makes angle α with the tangent (direction of u_1). Its tangential (whirl) part and radial (flow) part are

$$V_{w1} = V_1 \cos\alpha, \qquad V_{f1} = V_1 \sin\alpha$$

Step 2. Relative velocity at inlet. $\vec{V}_{r1} = \vec{V}_1 - \vec{u}_1$. The flow component does not change, and the tangential component becomes $V_{w1} - u_1$.

Step 3. Vane angle. The vane angle θ is the angle between $\vec{V}_{r1}$ and the direction of u_1:

$$\tan\theta = \frac{V_{f1}}{V_{w1} - u_1}$$

Step 4. Three runner types. If $V_{w1} > u_1$, the denominator is positive and $\theta < 90^\circ$. If $V_{w1} = u_1$, the denominator is zero, $\tan\theta \to \infty$, and $\theta = 90^\circ$. If $V_{w1} < u_1$, the tangential part of $\vec{V}_{r1}$ points backwards; the angle measured from u_1 is $\theta > 90^\circ$, and the acute angle on the other side has $\tan\,(180^\circ - \theta) = V_{f1}/(u_1 - V_{w1})$.

Step 5. Outlet triangle. With $V_{w2} = 0$, the absolute velocity is purely radial: $V_2 = V_{f2}$. The relative velocity is $\vec{V}_{r2} = \vec{V}_2 - \vec{u}_2$. Its flow component is V_{f2} and its tangential part is $-u_2$ (backwards), so the angle ϕ it makes with the reverse direction of u_2 satisfies

$$\tan\phi = \frac{V_{f2}}{u_2}$$

Result.

$$V_{w1} = V_1\cos\alpha, \quad V_{f1} = V_1\sin\alpha, \quad \tan\theta = \frac{V_{f1}}{V_{w1} - u_1}, \quad \tan\phi = \frac{V_{f2}}{u_2}$$

What it tells us. A larger guide-vane angle α gives more flow and less whirl. The runner vane angle at the inlet follows from α, V_1 and u_1, so the blades can be shaped for a smooth entry.

4.3 Work, efficiency and discharge CORE

IN SIMPLE WORDS

Work per newton of water is the whirl speed times the blade speed, divided by g. Efficiency compares that work with the head H. For the discharge, think of water crossing a cylinder: area is circumference times width, and flow is area times radial velocity. The same flow crosses the inlet cylinder and the outlet cylinder, because water cannot be created or lost.

$$\frac{\text{Work}}{\text{unit weight}} = \frac{V_{w1}u_1}{g}, \qquad \eta_h = \frac{V_{w1}u_1}{gH}, \qquad Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$$

B_1, B_2 = runner widths at inlet and outlet. If vane thickness t and number n are given, use $(\pi D_1 - nt)\,B_1$ in place of $\pi D_1 B_1$.

Design ratio	Definition	Typical range
Speed ratio K_u	$u_1/\sqrt{2gH}$	0.6–0.9
Flow ratio K_f	$V_{f1}/\sqrt{2gH}$	0.15–0.30
Breadth ratio n	B_1/D_1	0.10–0.45
Diameter ratio	D_2/D_1	about 0.5

Derivation from scratch: Francis work done, hydraulic efficiency, discharge, speed ratio and flow ratio

Aim. Prove $\text{work/weight} = V_{w1}u_1/g$, $\eta_h = V_{w1}u_1/(gH)$ and $Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$, and explain K_u and K_f.

Assumptions.

1. Euler's equation holds (Section 2.3).
2. Radial discharge, $V_{w2} = 0$.
3. Water is incompressible and flow is steady, so the same discharge crosses the inlet and outlet surfaces.
4. Velocity of flow is uniform around the runner.

Step 1. Euler's equation. The work done per unit weight is

$$w = \frac{1}{g}(V_{w1}u_1 \pm V_{w2}u_2)$$

Step 2. Set $V_{w2} = 0$. The second term is zero:

$$w = \frac{V_{w1}u_1}{g}$$

Step 3. Hydraulic efficiency. Divide the work per unit weight by the energy available per unit weight, which is the net head H:

$$\eta_h = \frac{w}{H} = \frac{V_{w1}u_1}{gH}$$

Step 4. Flow area at the inlet. Water crosses a cylindrical surface of diameter D_1 and width B_1. The surface area is circumference times width:

$$A_1 = \pi D_1 B_1$$

The velocity normal to this surface is the flow component V_{f1}.

Step 5. Continuity. Discharge equals area times normal velocity. At the inlet and the outlet:

$$Q = \pi D_1 B_1 V_{f1}, \qquad Q = \pi D_2 B_2 V_{f2}$$

Because the same Q passes both, $\pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$.

Step 6. Vane thickness. If n vanes of thickness t block part of the circumference, the free length is $(\pi D_1 - nt)$. Use $(\pi D_1 - nt)\,B_1$ in place of $\pi D_1 B_1$.

Step 7. Speed ratio and flow ratio. Free-fall speed from the head H is $\sqrt{2gH}$. Compare the blade speed and the flow speed with it:

$$K_u = \frac{u_1}{\sqrt{2gH}}, \qquad K_f = \frac{V_{f1}}{\sqrt{2gH}}$$

So $u_1 = K_u\sqrt{2gH}$ and $V_{f1} = K_f\sqrt{2gH}$. The breadth ratio $n = B_1/D_1$ is the width compared with the diameter.

Result.

$$\frac{\text{Work}}{\text{unit weight}} = \frac{V_{w1}u_1}{g}, \qquad \eta_h = \frac{V_{w1}u_1}{gH}, \qquad Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$$

What it tells us. Work grows with both the whirl speed and the blade speed. A wide, small-diameter runner gives the same discharge as a narrow, large-diameter one, so the shape (breadth ratio) is a design choice. Since $D_2 < D_1$, the outlet needs a larger width B_2 or a larger V_{f2} to carry the same Q.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

4.4 Draft tube ADVANCED

The draft tube lets the turbine be set above tailrace level without losing that height, and recovers most of the exit kinetic energy $V_2^2/(2g)$ as pressure. Both effects create a pressure below atmospheric at the runner exit, which raises the effective head on the runner.

Applying Bernoulli between runner exit (2) and draft tube outlet (3), with H_s = height of runner exit above tailrace and H_a = atmospheric pressure head:

$$\frac{p_2}{\rho g} = H_a - H_s - \left(\frac{V_2^2 - V_3^2}{2g} - h_f\right), \qquad \eta_d = \frac{\left(V_2^2 - V_3^2\right)/2g - h_f}{V_2^2/2g}$$

Types: conical (η_d up to about 90%), simple elbow (about 60%), elbow with circular inlet and rectangular outlet, Moody spreading tube. The cone half-angle is kept below about 8° to avoid flow separation.

Worked example

Step-by-step method.

1. Convert units: rpm, kW, m, and the percentage losses to decimals.
2. Find $\sqrt{2gH}$, then $u_1 = K_u\sqrt{2gH}$ and $V_{f1} = K_f\sqrt{2gH}$ if ratios are given.
3. Get V_{w1} from $\eta_h = V_{w1}u_1/(gH)$ (radial discharge, $V_{w2} = 0$).
4. Find the guide vane angle $\tan\alpha = V_{f1}/V_{w1}$ and the runner vane angle $\tan\theta = V_{f1}/(V_{w1} - u_1)$.
5. Find $D_1 = 60u_1/(\pi N)$.
6. Find $Q = P/\left(\rho g H \eta_o\right)$, then the width from $B_1 = Q/(\pi D_1 V_{f1})$.
7. Check each angle is sensible: α is acute, and θ is acute unless $V_{w1} < u_1$.

Example 4.0 (warm-up) CORE In a Francis turbine the water enters with $V_1 = 20\ \text{m/s}$ at guide vane angle $\alpha = 30^\circ$. The runner inlet speed is $u_1 = 12\ \text{m/s}$ and the discharge is radial. Find the runner vane angle at inlet and the work done per unit weight.

1. $V_{w1} = 20\cos 30^\circ = 17.32\ \text{m/s}$; $V_{f1} = 20\sin 30^\circ = 10\ \text{m/s}$
2. $\tan\theta = \dfrac{V_{f1}}{V_{w1} - u_1} = \dfrac{10}{17.32 - 12} = 1.880$, so $\theta = 62.0^\circ$
3. $\text{Work per unit weight} = \dfrac{V_{w1}u_1}{g} = \dfrac{17.32 \times 12}{9.81} = 21.19\ \text{N·m/N}$

Example 4.1 HOTS A Francis turbine with overall efficiency 85% produces 4000 kW under a head of 80 m. Peripheral velocity $= 0.65\sqrt{2gH}$, velocity of flow at inlet $= 0.22\sqrt{2gH}$, speed = 500 rpm, hydraulic losses = 10% of available energy. Assume radial discharge. Find the guide blade angle, runner vane angle at inlet, inlet diameter and width.

1. $\sqrt{2gH} = \sqrt{2 \times 9.81 \times 80} = 39.62\ \text{m/s}$; $u_1 = 0.65 \times 39.62 = 25.75\ \text{m/s}$; $V_{f1} = 0.22 \times 39.62 = 8.72\ \text{m/s}$
2. $\eta_h = 1 - 0.10 = 0.90 = \dfrac{V_{w1}u_1}{gH}$, so $V_{w1} = \dfrac{0.90 \times 9.81 \times 80}{25.75} = 27.43\ \text{m/s}$
3. $\tan\alpha = \dfrac{V_{f1}}{V_{w1}} = \dfrac{8.72}{27.43}$, so $\alpha = 17.6^\circ$
4. $\tan\theta = \dfrac{V_{f1}}{V_{w1} - u_1} = \dfrac{8.72}{1.68}$, so $\theta = 79.1^\circ$
5. $D_1 = \dfrac{60u_1}{\pi N} = \dfrac{60 \times 25.75}{\pi \times 500} = 0.984\ \text{m}$
6. $Q = \dfrac{P}{\rho g H \eta_o} = \dfrac{4\,000\,000}{1000 \times 9.81 \times 80 \times 0.85} = 5.996\ \text{m}^3/\text{s}$; $B_1 = \dfrac{Q}{\pi D_1 V_{f1}} = \dfrac{5.996}{\pi \times 0.984 \times 8.72} = 0.222\ \text{m}$

Common mistakes and exam tips.

- Using one blade speed for both ends. In a Francis runner $D_1 \neq D_2$, so $u_1 \neq u_2$.
- Forgetting that radial discharge means $V_{w2} = 0$ and $V_2 = V_{f2}$.
- Using V_1 or V_{r1} in $Q = \pi D_1 B_1 V_{f1}$. Only the flow (radial) component crosses the inlet area.
- Writing $\tan\theta = V_{f1}/(V_{w1} + u_1)$. The denominator is $V_{w1} - u_1$. When $V_{w1} < u_1$, the angle is obtuse.
- Putting the shaft power into $\rho g Q H$ directly. Use $Q = P/\left(\rho g H \eta_o\right)$ when P is the shaft output.
- Leaving out vane thickness when it is given: use $(\pi D_1 - nt)\,B_1$.

Check your understanding.

1. CORE Why is $u_1 \neq u_2$ in a Francis turbine? *(Inlet and outlet diameters differ, and $u = \pi DN/60$.)*
2. CORE Find Q for $D_1 = 0.5\ \text{m}$, $B_1 = 0.1\ \text{m}$, $V_{f1} = 4\ \text{m/s}$. *($Q = \pi \times 0.5 \times 0.1 \times 4 = 0.628\ \text{m}^3/\text{s}$.)*

3. HOTS If $V_{w1} = u_1$, what is the runner vane angle θ, and what kind of runner is it? *($\tan\theta \to \infty$, so $\theta = 90^\circ$: a medium-speed runner.)*

5. Kaplan turbine — axial-flow reaction turbine (L7) CORE

The Kaplan turbine is an axial-flow reaction turbine for low heads (below about 60 m) and large discharges. Its runner blades are adjustable, so it keeps high efficiency over a wide load range.

> **IN SIMPLE WORDS**
> A Kaplan turbine looks like a ship's propeller inside a pipe. Water enters the casing, is given a spin by guide vanes, turns through 90° and flows along the shaft past the blades. It suits big rivers with small height drops. The blades can twist, as an aeroplane propeller does, so the turbine stays efficient when the flow changes. A propeller turbine has the same shape but fixed blades.

5.1 Constructional details CORE

> **IN SIMPLE WORDS**
> Most parts are the same as in a Francis turbine. The new part is the hub with swivelling blades, moved by a servomotor. The space between the guide vanes and the blades is where the flow turns from radial to axial. The draft tube is vital because, at low head, a big share of the energy is still in the leaving water.

1. **Scroll casing, stay vanes and guide vanes** — as in the Francis turbine. Guide vanes give the water whirl.
2. **Whirl space** — between the guide vanes and the runner, where flow turns through 90° from radial to axial.
3. **Runner** — a large hub (boss) carrying 3–8 (usually 4–6) propeller-type blades. The blades swivel on their axes, operated by a servomotor inside the hollow shaft.
4. **Draft tube** — essential, since exit kinetic energy is a large share of the low head.

Propeller turbine = same machine with fixed blades; efficient only near design load.
Kaplan = adjustable blades plus adjustable guide vanes (double regulation), giving a flat part-load efficiency curve.

5.2 Velocity triangles and relations CORE

> **IN SIMPLE WORDS**
> In an axial machine the water goes in and out at the same distance from the shaft. So the blade speed is the same at the inlet and the outlet. The flow speed is also the same, because the flow area does not change. The flow passes through a ring-shaped area between the hub and the blade tip. Discharge is that ring area times the axial flow speed. The triangle rules are the same as for the Francis turbine.

Flow enters and leaves the runner at the same radius, so the blade velocity is the same at inlet and outlet. Calculations are done at the outer tip (diameter D_o) unless asked otherwise.

$$u_1 = u_2 = \frac{\pi D_o N}{60}, \qquad V_{f1} = V_{f2}, \qquad Q = \frac{\pi}{4}\left(D_o^2 - D_b^2\right) V_{f1}$$

D_b = hub (boss) diameter, $D_b/D_o \approx 0.35$–0.6. With axial discharge ($V_{w2} = 0$), the Francis relations apply:

$$\tan\alpha = \frac{V_{f1}}{V_{w1}}, \quad \tan\theta = \frac{V_{f1}}{V_{w1} - u_1}, \quad \tan\phi = \frac{V_{f2}}{u_2}, \quad \eta_h = \frac{V_{w1} u_1}{gH}$$

Derivation from scratch: Kaplan runner discharge and blade speed

Aim. Prove $Q = \frac{\pi}{4}\left(D_o^2 - D_b^2\right) V_{f1}$, and $u_1 = u_2$, $V_{f1} = V_{f2}$.

Assumptions.

1. Steady, incompressible flow, uniform axial velocity over the blade annulus.
2. The runner has outer (tip) diameter D_o and hub (boss) diameter D_b.
3. Water enters and leaves at the same radius, since the flow only turns from radial to axial in the whirl space and passes straight through the runner.

Step 1. Flow area. Water passes through the ring between the hub and the tip. The area of a circle of diameter D_o is $\frac{\pi}{4}D_o^2$ and the hub blocks a circle of diameter D_b:

$$A = \frac{\pi}{4}D_o^2 - \frac{\pi}{4}D_b^2 = \frac{\pi}{4}\left(D_o^2 - D_b^2\right)$$

Step 2. Continuity. Discharge is area times the velocity normal to it, which for axial flow is the flow component V_{f1}:

$$Q = A\,V_{f1} = \frac{\pi}{4}\left(D_o^2 - D_b^2\right) V_{f1}$$

Step 3. Same flow speed at outlet. The same area is available at the outlet, so continuity gives $AV_{f1} = AV_{f2}$, hence

$$V_{f1} = V_{f2}$$

Step 4. Same blade speed. Both inlet and outlet are at the same radius, so the blade speed is the same:

$$u_1 = u_2 = \frac{\pi D N}{60}$$

with $D = D_o$ at the tip. The Euler work $V_{w1}u_1 - V_{w2}u_2$ with $V_{w2} = 0$ then gives $V_{w1}u_1/g$.

Result.

$$Q = \frac{\pi}{4}\left(D_o^2 - D_b^2\right) V_{f1}, \qquad u_1 = u_2 = \frac{\pi D_o N}{60}, \qquad V_{f1} = V_{f2}$$

What it tells us. The hub takes up area, so a bigger hub needs a faster axial flow for the same discharge. At the tip, the blade speed is greatest. In practice the blades are twisted to follow the different speeds at different radii.

Feature	Francis	Kaplan
Flow through runner	Radial in, axial out (mixed)	Axial
Head range	Medium, about 60–250 m	Low, below about 60 m
Number of blades	16–24, fixed	3–8, adjustable
u at inlet and outlet	Different ($D_1 \neq D_2$)	Same
Part-load efficiency	Falls noticeably	Stays high
Specific speed (SI)	51–255	255–860

Worked example

Step-by-step method.

1. Find the discharge: $Q = P/\left(\rho g H \eta_o\right)$.
2. Find the flow area: $\frac{\pi}{4}\left(D_o^2 - D_b^2\right)$, then $V_{f1} = Q/\text{area}$.
3. Get V_{w1} from the guide blade angle: $V_{w1} = V_{f1}/\tan\alpha$.
4. Find u_1 from $\eta_h = V_{w1}u_1/\left(gH\right)$.
5. Runner angles: $\tan\theta = V_{f1}/\left(V_{w1} - u_1\right)$ and $\tan\phi = V_{f1}/u_1$ (because $u_2 = u_1$ and $V_{f2} = V_{f1}$).
6. Speed: $N = 60u_1/\left(\pi D_o\right)$.

Example 5.0 (warm-up) CORE A Kaplan runner has outer diameter $2\ \text{m}$, hub diameter $1\ \text{m}$ and axial flow speed $5\ \text{m/s}$. It runs at $100\ \text{rpm}$. Find the discharge and the blade speed at the tip.

1. Flow area $= \frac{\pi}{4}\left(2^2 - 1^2\right) = 2.356\ \text{m}^2$
2. $Q = A\,V_{f1} = 2.356 \times 5 = 11.78\ \text{m}^3/\text{s}$
3. $u_1 = \frac{\pi D_o N}{60} = \frac{\pi \times 2 \times 100}{60} = 10.47\ \text{m/s}$

Example 5.1 CORE A Kaplan turbine under a head of 20 m develops 17 500 kW shaft power. Outer runner diameter = 4.0 m, hub diameter = 1.6 m, guide blade angle at the outer edge = 60°. Hydraulic and overall efficiencies are 90% and 85%. Whirl at outlet is zero. Find the runner vane angles at the outer edge and the speed.

1. $Q = \frac{P}{\rho g H \eta_o} = \frac{17\,500\,000}{1000 \times 9.81 \times 20 \times 0.85} = 104.93\ \text{m}^3/\text{s}$
2. Flow area $= \frac{\pi}{4}\left(4.0^2 - 1.6^2\right) = 10.556\ \text{m}^2$; $V_{f1} = 104.93/10.556 = 9.94\ \text{m/s}$
3. $V_{w1} = V_{f1}/\tan 60^\circ = 5.74\ \text{m/s}$
4. $\eta_h = V_{w1}u_1/\left(gH\right)$: $u_1 = \frac{0.90 \times 9.81 \times 20}{5.74} = 30.76\ \text{m/s}$

5. $V_{w1} < u_1$, so θ is obtuse: $\tan\left(180^\circ - \theta\right) = \dfrac{V_{f1}}{u_1 - V_{w1}} = \dfrac{9.94}{25.02}$, so $180^\circ - \theta = 21.7^\circ$ and $\theta = 158.3^\circ$; $\tan\phi = 9.94/30.76$, so $\phi = 17.9^\circ$
6. $N = \dfrac{60u}{\pi D_o} = \dfrac{60 \times 30.76}{\pi \times 4.0} = 146.9\,\text{rpm}$

Common mistakes and exam tips.

- Using the full circle $\frac{\pi}{4}D_o^2$ for the flow area. Subtract the hub: $\frac{\pi}{4}\left(D_o^2 - D_b^2\right)$.
- Using different u at inlet and outlet. In a Kaplan runner $u_1 = u_2$ at a given radius.
- Using the hub diameter for blade speed. Use D_o at the tip unless the question gives another radius.
- Mixing up diameter and radius in $u = \pi DN/60$.
- Using the shaft power in Q without η_o, or using η_h where η_o belongs.

Check your understanding.

1. CORE A runner has $D_o = 2\,\text{m}$, $D_b = 1\,\text{m}$, $V_f = 5\,\text{m/s}$. Find Q. *($Q = \frac{\pi}{4}\left(4 - 1\right) \times 5 = 11.78\,\text{m}^3/\text{s}$.)*
2. HOTS Why does a Kaplan turbine keep high efficiency at part load? *(Both the runner blades and the guide vanes can be turned, so the angles still suit the water as discharge changes.)*
3. CORE Why is $u_1 = u_2$ in a Kaplan runner? *(Water enters and leaves at the same radius.)*

6. Specific speed, unit quantities and selection of turbines (L8) CORE

Specific speed is the single number that decides the turbine type for a site: low N_s means Pelton, medium means Francis, high means Kaplan.

IN SIMPLE WORDS

Turbines come in many sizes, so we need one number that describes the shape of a runner, not its size. Specific speed is that number. Imagine shrinking or enlarging a turbine until it gives 1 kW under 1 m head, and then ask how fast it spins. A slow, high-head machine has a small N_s and a fast, low-head machine has a large N_s. Once you know N_s from the site, you can pick the turbine.

6.1 Specific speed CORE

IN SIMPLE WORDS

N is the actual speed in rpm, P is the power in kW and H is the net head in m. The formula puts speed up and head down: more speed and power raise N_s, more head lowers it. Always use kW and rpm. Other units give a different number.

Specific speed N_s is the speed of a geometrically similar turbine that develops 1 kW under a head of 1 m. From $u \propto \sqrt{H}$, $D \propto \sqrt{H}/N$ and $P \propto QH \propto D^2 H^{3/2}$:

$$N_s = \frac{N\sqrt{P}}{H^{5/4}} \qquad (N \text{ in rpm},\ P \text{ in kW},\ H \text{ in m})$$

In MKS units (P in metric hp), values are about 1.16 times the SI (kW) values. State the unit used in every answer.

Derivation from scratch: specific speed of a turbine

Aim. Prove $N_s = N\sqrt{P}/H^{5/4}$.

Assumptions.

1. The turbines compared are geometrically similar, with similar velocity triangles and the same efficiency.
2. D is any one characteristic diameter of the runner.
3. N_s is defined as the speed N at which a turbine of this shape would develop $P = 1\,\text{kW}$ under $H = 1\,\text{m}$.

Step 1. Blade speed and head. Blade speed is proportional to the speed gained by a free fall through H (the speed ratio K_u is the same for similar turbines):

$$u \propto \sqrt{H}$$

Step 2. Blade speed from rpm. From $u = \pi DN/60$ we have $u \propto DN$. Equate the two proportionalities, $DN \propto \sqrt{H}$, so

$$D \propto \frac{\sqrt{H}}{N}$$

Step 3. Discharge. Discharge is flow area times flow velocity. The area is proportional to D^2, and the flow velocity is proportional to $\sqrt{H}$ (the flow ratio K_f is the same):

$$Q \propto D^2\sqrt{H}$$

Step 4. Power. Power is $P = \eta\rho g Q H$, with η constant, so

$$P \propto QH \propto D^2\sqrt{H}\cdot H = D^2 H^{3/2}$$

Step 5. Eliminate D. Substitute $D^2 \propto H/N^2$ from Step 2:

$$P \propto \frac{H}{N^2}\,H^{3/2} = \frac{H^{5/2}}{N^2}$$

Step 6. Insert a constant. Write $P = C\,H^{5/2}/N^2$ with C the same for all similar turbines.

Step 7. Apply the definition. When $P = 1$ and $H = 1$, $N = N_s$:

$$1 = \frac{C}{N_s^2} \quad \Rightarrow \quad C = N_s^2$$

Step 8. Solve for N_s. Put $C = N_s^2$ back:

$$P = \frac{N_s^2\, H^{5/2}}{N^2} \quad \Rightarrow \quad N_s^2 = \frac{N^2 P}{H^{5/2}} \quad \Rightarrow \quad N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

Result.

$$N_s = \frac{N\sqrt{P}}{H^{5/4}}$$

What it tells us. N_s depends on the shape of the runner only, so turbines with the same N_s are similar in design. Low N_s (high head, small flow) is Pelton; high N_s (low head, large flow) is Kaplan. Example: $N = 300$, $P = 5000$, $H = 300$ gives $N_s = 17.0$, in the Pelton range.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

6.2 Unit quantities ADVANCED

Unit quantities give the performance of the same turbine under a head of 1 m. They are used to predict performance when the head at a site changes.

Unit quantity	Definition	Use: same turbine at a new head
Unit speed N_u	$N/\sqrt{H}$	$N_2 = N_1\sqrt{H_2/H_1}$
Unit discharge Q_u	$Q/\sqrt{H}$	$Q_2 = Q_1\sqrt{H_2/H_1}$
Unit power P_u	$P/H^{3/2}$	$P_2 = P_1\,(H_2/H_1)^{3/2}$

6.3 Selection of water turbines HOTS

IN SIMPLE WORDS

Choose a turbine by looking at the head and the flow of the site. High head and small flow: Pelton. Medium head and flow: Francis. Low head and a large flow: Kaplan. Then check part-load behaviour, cavitation, silt and cost. The table below gives the ranges of N_s.

Turbine	Net head	Specific speed N_s (SI)	Flow
Pelton, single jet	Above 250 m	8.5–30	Tangential
Pelton, two or more jets	Above 250 m	30–51	Tangential
Francis	60–250 m	51–255	Mixed (radial-axial)
Kaplan / propeller	Below 60 m	255–860	Axial

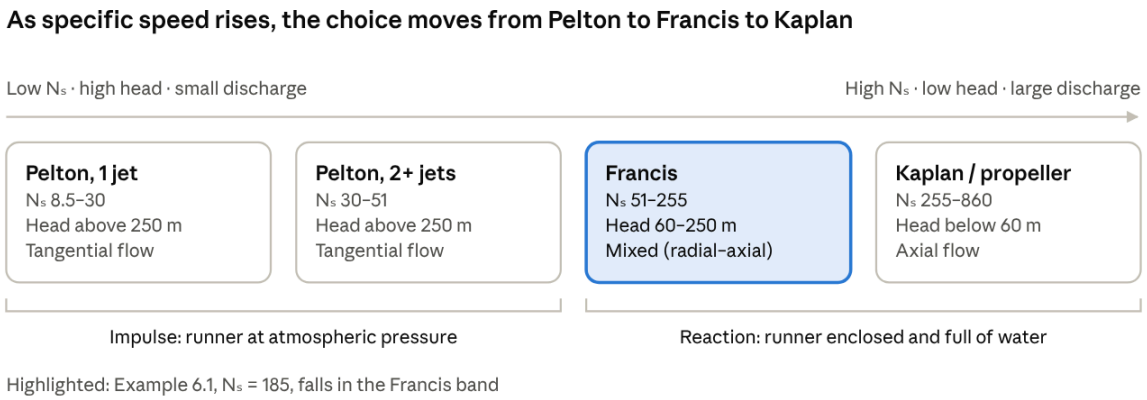


Figure 4. Turbine selection map by specific speed and head

Head ranges overlap in practice (Francis units run from about 30 m to over 500 m). Other selection factors:

- **Part-load efficiency** — Pelton and Kaplan stay efficient at part load; Francis and propeller turbines drop off.
- **Cavitation** — high-N_s reaction turbines need a low setting (small H_s) to avoid cavitation.
- **Runaway speed and generator speed** — a higher N_s allows higher rotational speed for a given head, so a smaller, cheaper generator.
- **Silt in water** — Pelton buckets tolerate and are easier to repair than reaction runners.
- **Installation and civil cost** — Pelton needs no draft tube; Kaplan needs large casings and draft tubes.

Worked examples

Step-by-step method.

1. Write N in rpm, P in kW, H in m.
2. Compute $\sqrt{P}$ and $H^{5/4}$ separately with a calculator ($H^{5/4} = H\sqrt[4]{H}$).
3. Divide: $N_s = N\sqrt{P}/H^{5/4}$.
4. Compare with the table in 6.3 and name the turbine.
5. State the unit system used for N_s.

Example 6.0 (warm-up) CORE A turbine develops $10\,000\ \text{kW}$ at $300\ \text{rpm}$ under a head of $100\ \text{m}$. Find N_s and the turbine type.

1. $\sqrt{P} = \sqrt{10\,000} = 100$; $H^{5/4} = 100^{1.25} = 316.2$
2. $N_s = \dfrac{N\sqrt{P}}{H^{5/4}} = \dfrac{300 \times 100}{316.2} = 94.9$
3. 94.9 lies in 51–255, so a **Francis turbine**.

Example 6.1 HOTS A turbine develops 7500 kW under a head of 30 m at 150 rpm. Find N_s and suggest the type.

$N_s = \dfrac{150 \times \sqrt{7500}}{30^{1.25}} = \dfrac{150 \times 86.6}{70.2} = 185$, which lies in 51–255, so a **Francis turbine**.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

Example 6.2 ADVANCED A turbine develops 9000 kW at 450 rpm under a head of 100 m. Find its speed and power if the head falls to 81 m.

1. $N_2 = 450 \times \sqrt{81/100} = 405 \text{ rpm}$
2. $P_2 = 9000 \times \left(81/100\right)^{1.5} = 6561 \text{ kW}$

Common mistakes and exam tips.

- Using power in W or hp instead of kW (or mixing units between the answer and the table). State the unit.
- Using the shaft speed in rad/s. N must be in rpm.
- Computing $H^{5/4}$ wrongly, for example $H^{5/4} \neq H^{1.5}$. Check with $100^{1.25} = 316.2$.
- Using the gross head instead of the net head.
- Reading a type from N_s without checking the head. High head with a large N_s is a warning to recheck the arithmetic.

Check your understanding.

1. CORE In what units are N, P and H used in N_s? *(rpm, kW and m.)*
2. CORE A turbine has $N_s = 20$. Which type is it? *(A single-jet Pelton wheel, since $8.5 < 20 < 30$.)*
3. HOTS A site has a head of 15 m and a very large flow. Which turbine suits it and why? *(A Kaplan turbine: low head and large discharge give a high N_s in the 255–860 range.)*

7. Formula sheet and practice questions CORE

7.1 Quick formula sheet CORE

Topic	Result
Fixed flat plate, normal	$F = \rho a V^2$
Fixed symmetrical curved plate	$F = \rho a V^2 \left(1 + \cos\theta \right)$
Series of flat plates	$\eta = 2u \left(V - u \right) / V^2$, max 50% at $u = V/2$
Euler equation (work per N)	$\left(V_{w1} u_1 \pm V_{w2} u_2 \right) / g$
Water power	$\rho g Q H / 1000$ kW
Pelton jet velocity	$V_1 = C_v \sqrt{2gH}$
Pelton hydraulic efficiency	$\eta_h = 2 \left(V_1 - u \right) \left(1 + k \cos\phi \right) u / V_1^2$
Pelton maximum efficiency	$u = V_1/2$, $\eta_{h,\ \max} = \left(1 + k \cos\phi \right) / 2$
Pelton bucket number	$z = 15 + D / \left(2d \right)$
Francis discharge	$Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$
Kaplan discharge	$Q = \left(\pi/4 \right) \left(D_o^2 - D_b^2 \right) V_{f1}$
Radial / axial discharge	$\eta_h = V_{w1} u_1 / \left(gH \right)$
Specific speed	$N_s = N \sqrt{P} / H^{5/4}$
Unit quantities	$N_u = N / \sqrt{H}$, $Q_u = Q / \sqrt{H}$, $P_u = P / H^{3/2}$

7.2 Short-answer questions (2 marks) CORE

1. CORE Define hydraulic, mechanical and overall efficiency of a turbine.
2. CORE Distinguish between impulse and reaction turbines.
3. HOTS Why are Pelton buckets designed to deflect the jet through 160–165° and not 180°?
4. CORE What is the function of the splitter in a Pelton bucket?
5. CORE State the condition for maximum hydraulic efficiency of a Pelton wheel.
6. CORE Define specific speed. What is its significance?

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

7. ADVANCED What is a draft tube? Give two of its functions.

8. CORE Differentiate between a Kaplan and a propeller turbine.
9. CORE Define jet ratio and speed ratio.

Advanced (optional). Beyond what the AKTU exam usually asks; read it after the core material is secure.

10. ADVANCED What are unit quantities and why are they used?

7.3 Long-answer and numerical questions (10 marks) CORE

1. CORE Derive the expression for force exerted by a jet on a series of flat plates mounted on a wheel. Show that maximum efficiency is 50%.
2. CORE Derive the expression for work done by a jet on a moving unsymmetrical curved vane when the jet strikes tangentially at one tip.
3. CORE Draw the inlet and outlet velocity triangles of a Pelton wheel and derive the expression for maximum hydraulic efficiency.
4. CORE Explain with a neat sketch the oil-pressure governing of a Pelton wheel. Why is a deflector needed?
5. CORE Describe the construction of a Francis turbine with a labelled sketch, and draw its velocity triangles.
6. HOTS Compare Francis and Kaplan turbines. Explain why a Kaplan turbine has high part-load efficiency.
7. CORE Derive the expression for specific speed of a turbine. How is it used for turbine selection?
8. CORE A jet of water of diameter 50 mm moving at 25 m/s strikes a series of semicircular vanes moving at 10 m/s. Find the force, work done per second and efficiency. *(Ans: 1473 N, 14.7 kW, 96%)*
9. HOTS A Pelton wheel works under a head of 500 m with $C_v = 0.98$ and develops 4000 kW shaft power at 400 rpm. Overall efficiency = 85%, speed ratio = 0.46. Find the discharge, pitch-circle diameter, jet diameter and number of buckets.
10. CORE An inward-flow reaction turbine has inlet and outlet diameters 1.0 m and 0.5 m, runs at 300 rpm, guide blade angle 15° and constant velocity of flow 4 m/s. With radial discharge, find the runner vane angles at inlet and outlet and the work done per unit weight.
11. HOTS A Kaplan turbine develops 9000 kW under a head of 15 m. Overall efficiency = 88%, speed ratio = 2.0, flow ratio = 0.6, hub diameter = $0.35\times$ outer diameter. Find the runner diameter and speed. Calculate its specific speed.
12. HOTS A turbine is to develop 8000 kW at 200 rpm under 25 m head. Find the specific speed, select the type, and find the speed and power if the head becomes 16 m.